

CP 330 EDGE AI

Assignment 1: IMU-Based Human Activity Recognition using Nicla Vision

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1 Introduction

This report documents the end-to-end development of a Human Activity Recognition (HAR) system utilizing the Nicla Vision hardware platform. The primary objective of this assignment is to execute a complete TinyML pipeline, bridging the gap between raw sensor data acquisition and real-time edge deployment. The project workflow begins with the systematic collection of wrist mounted Inertial Measurement Unit (IMU) data, specifically 3-axis Accelerometer and Gyroscope readings sampled at 50 Hz across twelve distinct daily activities, ranging from static postures like sitting to dynamic movements such as jogging and cycling. Following data acquisition, a sliding window methodology is applied to extract critical time-domain features, which are then used to train and evaluate a machine learning model for activity classification. The final phase of the project demonstrates the model's practical viability through real-time inference, validating its ability to correctly identify user activities in a live environment.

1.1 System Overview

The core hardware platform utilized for this project is the **Arduino Nicla Vision** as shown in Fig. 1, a compact, low-power edge computing device optimized for TinyML applications. The data acquisition relies on the board's integrated 6-axis Inertial Measurement Unit (IMU), specifically utilizing:

- **3-Axis Accelerometer:** To measure proper acceleration and linear movement.
- **3-Axis Gyroscope:** To measure angular velocity and rotational orientation.

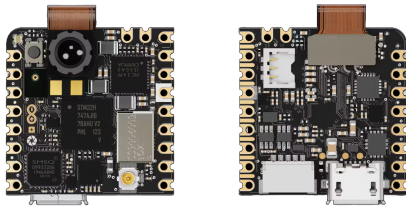


Figure 1: Nicla Vision Device view

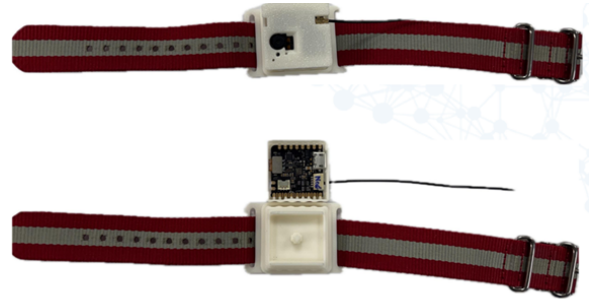


Figure 2: Nicla Band

Data is sampled at a frequency of **50 Hz**, providing sufficient temporal resolution to capture human motion dynamics while maintaining a manageable data load for embedded processing.

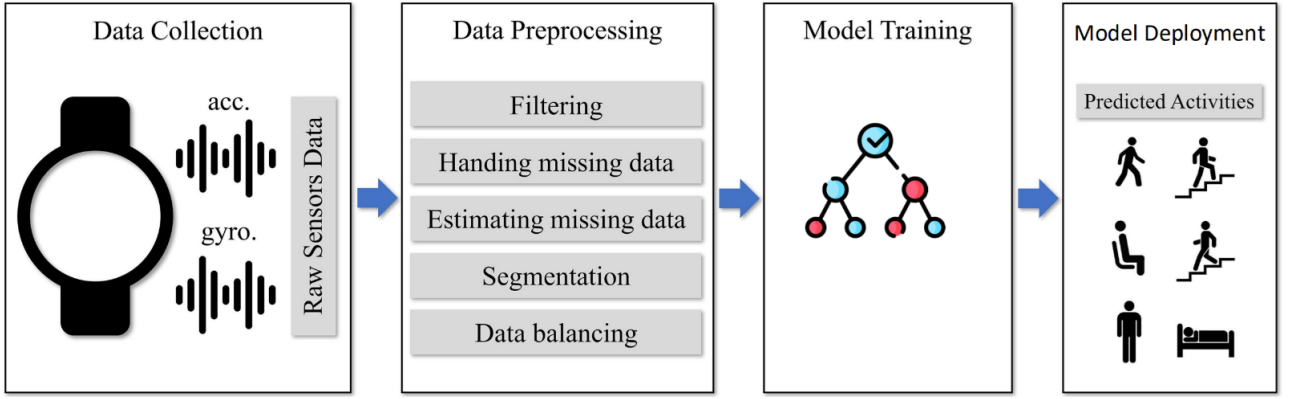


Figure 3: Overall workflow of the assignment

1.2 Scope of the assignment

This report documents the complete development lifecycle of the HAR system, structured into three distinct phases shown in Fig. 3:

1. **Data Collection:** The systematic gathering of raw sensor data for 12 activity classes. Strict protocols were followed to ensure dataset quality, collecting approximately 9,000 samples per activity (3 minutes duration) without applying on-sensor filtering at the collection stage.
2. **Feature Engineering & Model Development:** The transition from raw time-series data to interpretable features. This involves a sliding window approach (Window Size, Overlap) to segment the data. Statistical features (mean, max, standard deviation, etc.) are extracted from these windows to train a machine learning classifier, focusing on maximizing accuracy and minimizing confusion between similar activities.
3. **Real-Time Deployment:** The final phase involves porting the trained model back to the Nicla Vision board. This demonstrates the system’s ability to perform real-time inference, converting live sensor streams into immediate activity classifications in a “real-world” environment.

2 Data Collection Methodology

The data acquisition framework for this study utilized the Arduino Nicla Vision as the primary edge sensing device, specifically configured to capture high-fidelity motion data from a wrist-mounted position. The device was secured on the participant’s right wrist as shown in Fig. 2, with the data cable oriented towards the right to ensure consistent sensor axis alignment across all trials. The core sensing modality relied on the board’s integrated 6-axis Inertial Measurement Unit (IMU), which simultaneously recorded linear acceleration via a 3-axis accelerometer and

angular velocity via a 3-axis gyroscope. To capture the nuances of human motion—ranging from subtle fine-motor tasks to rapid locomotion—the sensor sampling frequency was standardized at 50 Hz. This configuration provided sufficient temporal resolution to distinguish activity patterns while maintaining a manageable data throughput for wireless transmission.

To facilitate robust data logging, a custom Python script (`receive_data.py`) was developed to execute on a host computer, serving as a data sink for the wireless sensor stream. The system utilized the User Datagram Protocol (UDP) for low-latency communication, with the host script binding to a local socket on port 5005 to listen for incoming sensor packets. Upon receiving the raw data stream, the script decoded the payload to extract the six IMU channels and appended a precise host-side timestamp to ensure temporal accuracy, disregarding the board’s internal timer to avoid synchronization drift. The data was subsequently serialized and stored in Comma-Separated Values (CSV) format, with a file structure explicitly defining columns for the timestamp, the three accelerometer axes, the three gyroscope axes, and the ground-truth activity label. This automated logging pipeline included a five-second countdown prior to initialization, ensuring the participant could settle into the specific posture before recording commenced.

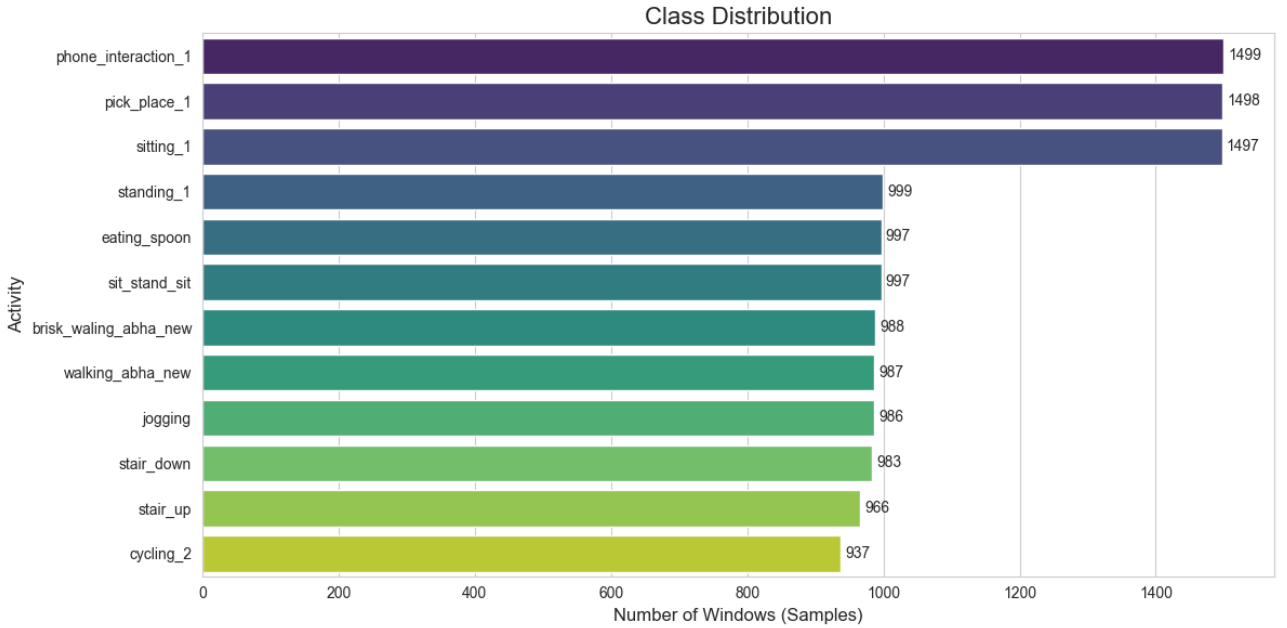


Figure 4: Class distribution of collected data

The experimental protocol encompassed the collection of twelve distinct activity classes representing a comprehensive range of daily human behavior shown in Fig. 4. These activities included static postures such as sitting and standing; locomotive movements including walking at a normal pace, brisk walking, jogging, cycling, and ascending or descending stairs; and complex transitional or functional tasks such as the *sit-stand-sit* sequence, phone interaction (typing/scrolling), eating with a spoon, and a pick-and-place operation. To meet the assignment requirement of approximately 9,000 samples per class (equating to three minutes of data),

the acquisition window in the software was effectively set between 200 and 300 seconds given Table 1. This extended duration provided a safety buffer to account for potential network instability or momentary device disconnections, ensuring that the final cleaned dataset for each activity strictly exceeded the minimum required sample count. Each activity was recorded in a dedicated session and saved as an individual file to prevent label noise and ensure data integrity for the subsequent feature engineering phase.

Table 1: Data Collection Summary

Activity	File Name	Samples	Columns
Brisk Walking	brisk_waling_abha_new_20260220_214337	~9,900	8
Cycling	cycling_20260125_155502.csv	~9,380	8
Eating with Spoon	eating_spoon_20260125_173356.csv	~9,980	8
Jogging	jogging_20260125_181023.csv	~9,870	8
Phone Interaction	phone_interaction_1_20260201_005353.csv	~15,000	8
Pick and Place	pick_place_1_20260131_114737.csv	~14,990	8
Sitting	sitting_1_20260131_112629.csv	~14,980	8
Sit-Stand-Sit	sit_stand_sit_20260125_182628.csv	~9,980	8
Stair Down	stair_down_20260125_181616.csv	~9,840	8
Stair Up	stair_up_20260125_182012.csv	~9,670	8
Standing	standing_1_20260201_001151.csv	~10,000	8
Walking	walking_abha_new_20260220_213943.csv	~9,882	8

2.1 Exploratory Data Analysis and Visualization

Exploratory Data Analysis (EDA) was conducted to validate signal integrity and assess class separability. Time-series visualizations of raw IMU streams confirmed distinct periodic waveforms for dynamic activities compared to the flat responses of static postures. Statistical distribution analysis, utilizing box plots and Kernel Density Estimation (KDE), identified feature outliers and highlighted potential overlaps between spectrally similar classes such as Standing and Phone Interaction. Furthermore, dimensionality inspection via t-SNE demonstrated clear clustering for distinct activities, while visually defining the boundary complexities inherent in transitional tasks.

Table 2: Sample Preview of Brisk Walking Dataset

Ax	Ay	Az	Gx	Gy	Gz	Activity
0.553711	-0.848389	-0.292480	90.75926	-92.95652	152.34380	brisk_waling
0.435791	-0.815308	-0.251465	87.28026	-66.71144	126.28172	brisk_waling
0.422241	-0.833496	-0.253296	81.05466	-30.94482	101.74560	brisk_waling
0.466797	-0.861938	-0.302124	69.45802	5.310058	84.96092	brisk_waling
0.512940	-0.849610	-0.361206	55.78612	35.52246	78.00294	brisk_waling

2.1.1 Brisk Walking

The data for brisk walking (Table 2) exhibits a highly periodic and consistent sinusoidal waveform, reflecting the rhythmic arm swing associated with rapid locomotion. The time-series plots show a high signal-to-noise ratio with clear peak-to-peak intervals. This is corroborated by the box plot analysis, which reveals a significantly wide Interquartile Range (IQR) on the Y-axis (vertical acceleration), confirming it as the primary axis of motion. Additionally, the gyroscope data displays numerous outliers ($\approx \pm 200$ dps), indicating sharp, rapid angular changes at the varying extremes of the swing phase, as shown in Fig. 5, 6.

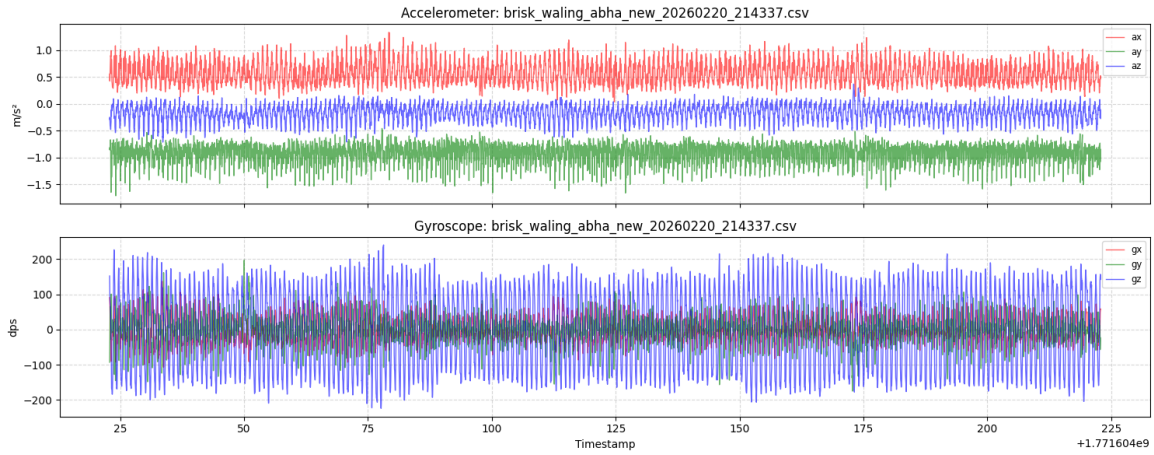


Figure 5: Brisk Walking

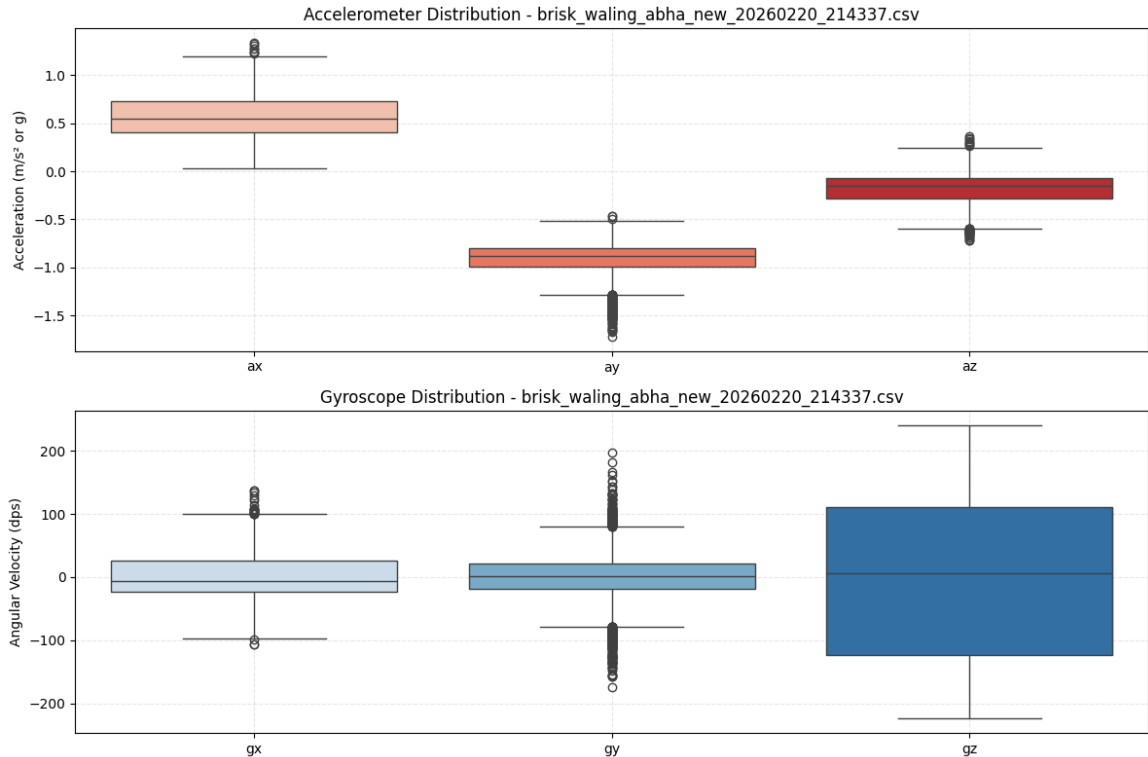


Figure 6: Brisk Walking

2.1.2 Cycling

In contrast to the large oscillations observed in walking, the cycling data presents a high-frequency, low-amplitude vibration pattern. The signal remains relatively stable around the mean, accurately reflecting the wrist's fixed position on the handlebars. The observed variance is primarily driven by road noise and minor steering corrections rather than active limb movement, resulting in a distinct *texture* in the time-domain signal that clusters tightly in the t-SNE projection, as shown in Fig. 7, 8.

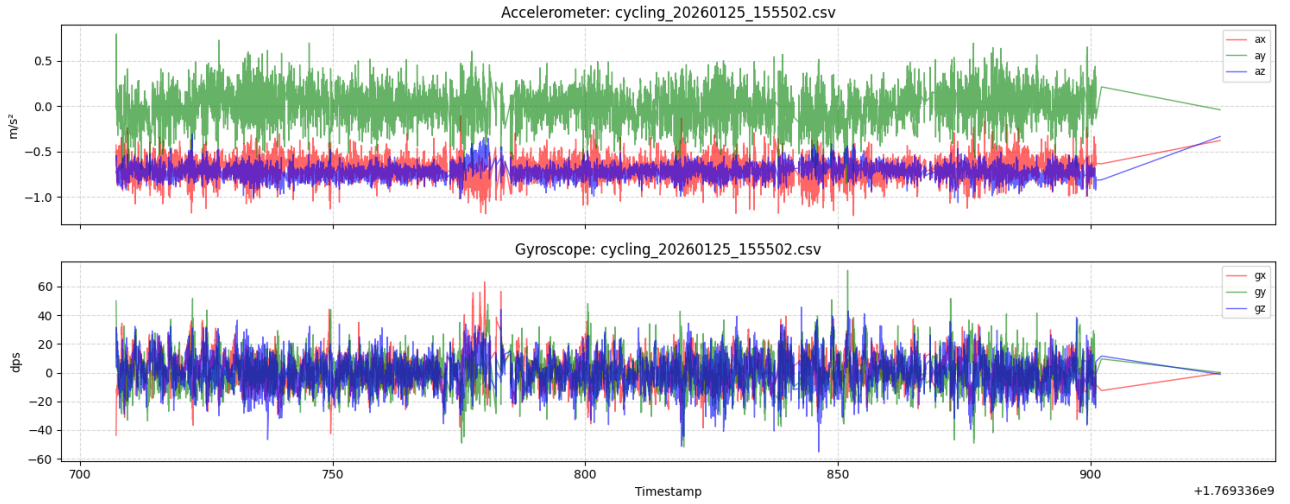


Figure 7: Cycling

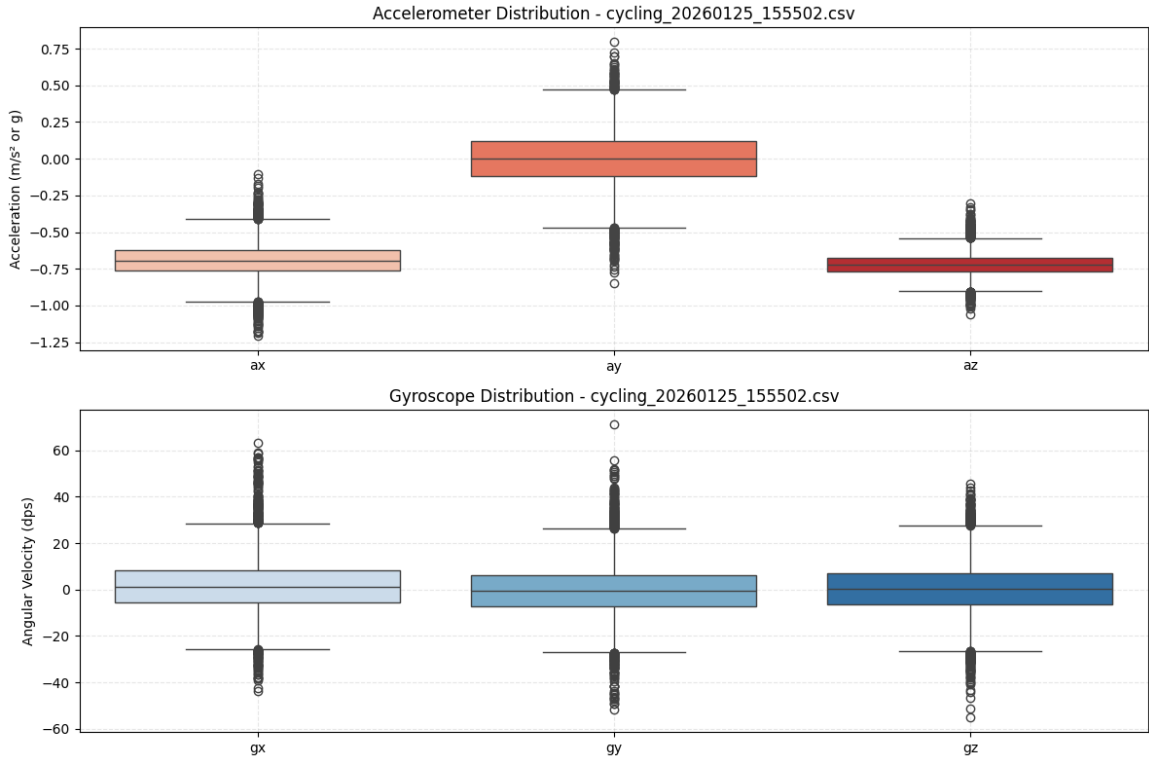


Figure 8: Cycling

2.1.3 Eating with Spoon

This activity displays a unique quasi-periodic signal structure characterized by distinct "bursts" of motion followed by brief stationary pauses. The accelerometer traces clearly delineate the specific functional cycle—scooping, lifting to the mouth, and lowering—creating a repetitive "hump" pattern. This structural irregularity makes it visually and statistically distinct from continuous locomotive activities, as the signal variance is episodic rather than constant, as shown in Fig. 9, 10.

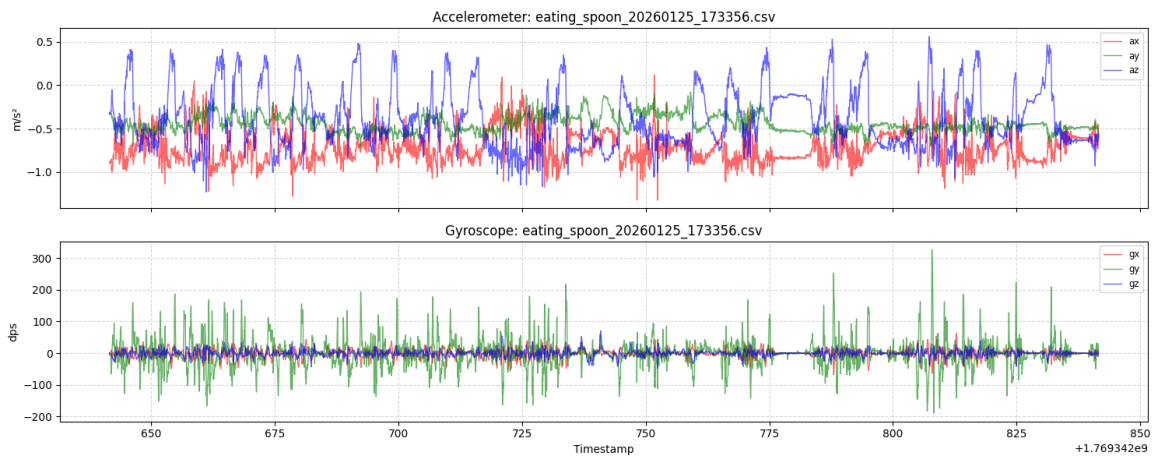


Figure 9: Eating with Spoon

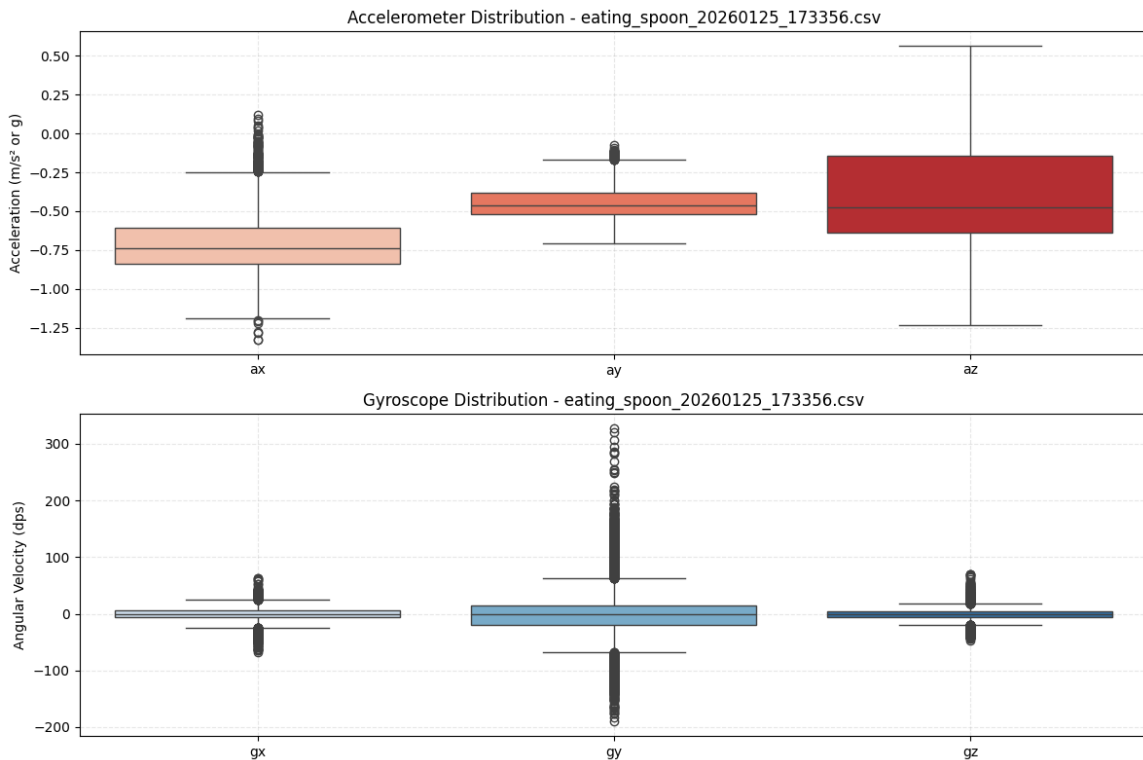


Figure 10: Eating with Spoon

2.1.4 Jogging

Jogging represents the most dynamic activity in the dataset, exhibiting the highest amplitude displacement and frequency density. The time-series data shows sharp, deep troughs and rapid oscillations, capturing the intense vertical ground reaction forces. The corresponding box plot analysis highlights extreme variance and significant outliers across all axes (with accelerometer readings frequently exceeding $\pm 2g$ and gyroscope values nearing saturation), indicative of the high-impact nature and rapid cadence of running, as shown in Fig. 11, 12.

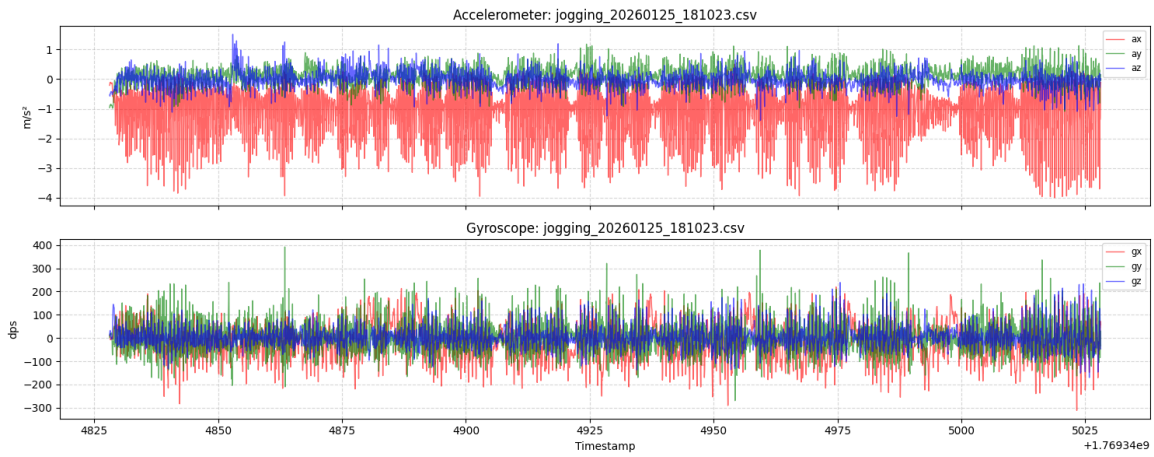


Figure 11: Jogging

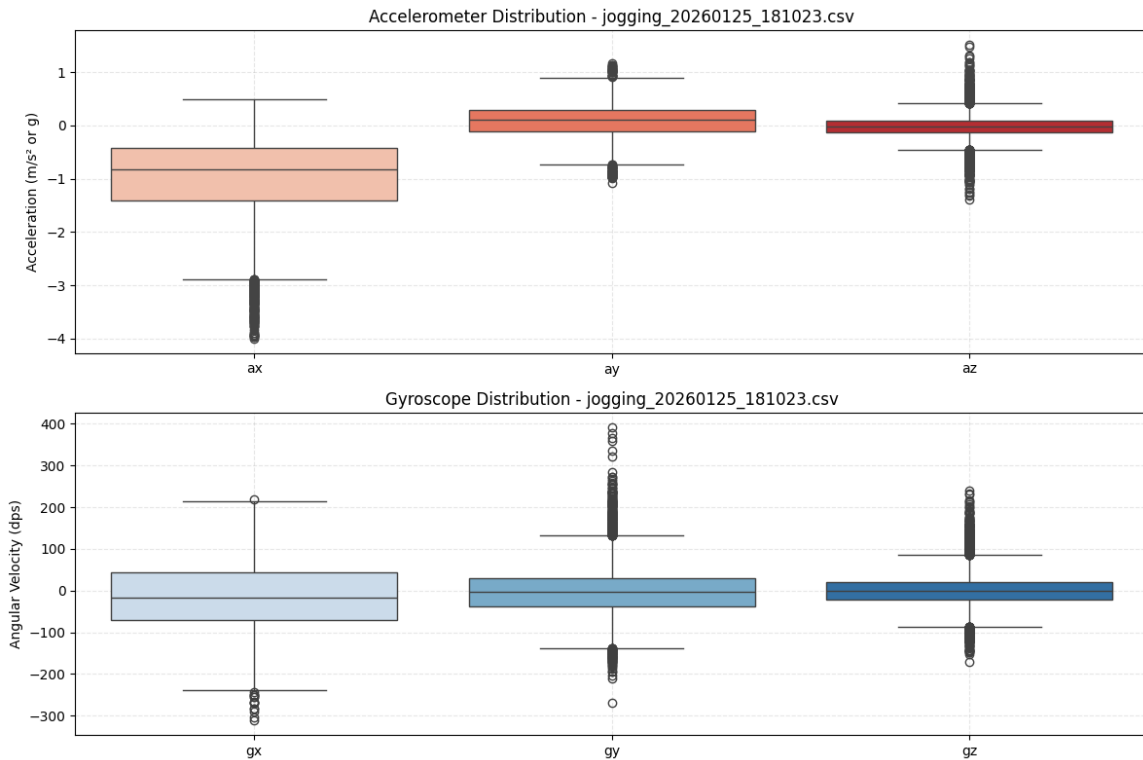


Figure 12: Jogging

2.1.5 Phone Interaction

Phone Interaction The signal profile for phone interaction is predominantly static but distinguishable from pure resting states by the presence of micro-tremors and minor orientation shifts. While the accelerometer data remains stable (reflecting the holding of the device), the gyroscope detects subtle, high-frequency spikes corresponding to thumb movements (typing/scrolling) and slight wrist adjustments. The box plots confirm this with a very narrow IQR but a higher density of outliers compared to the *Sitting* class, as shown in Fig. 13, 14.

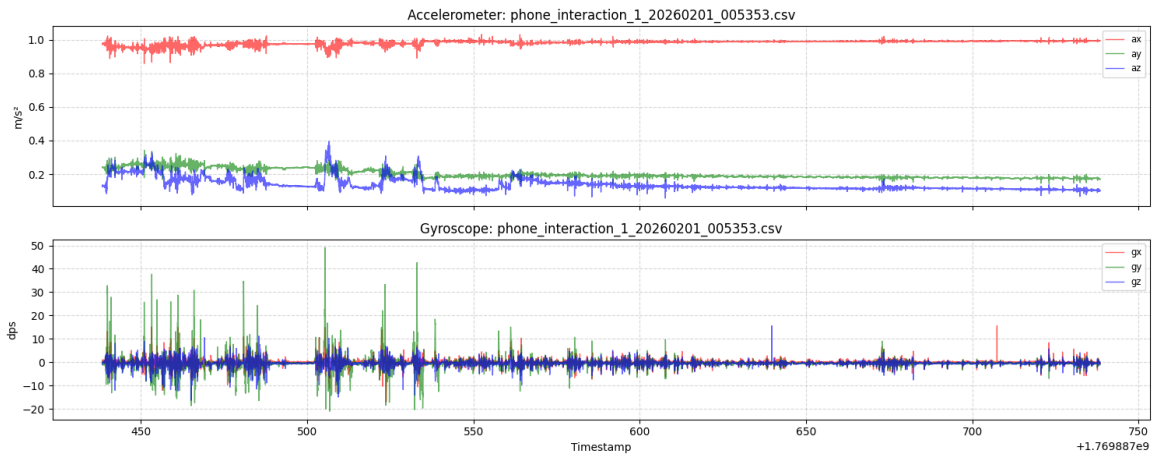


Figure 13: Phone Interaction

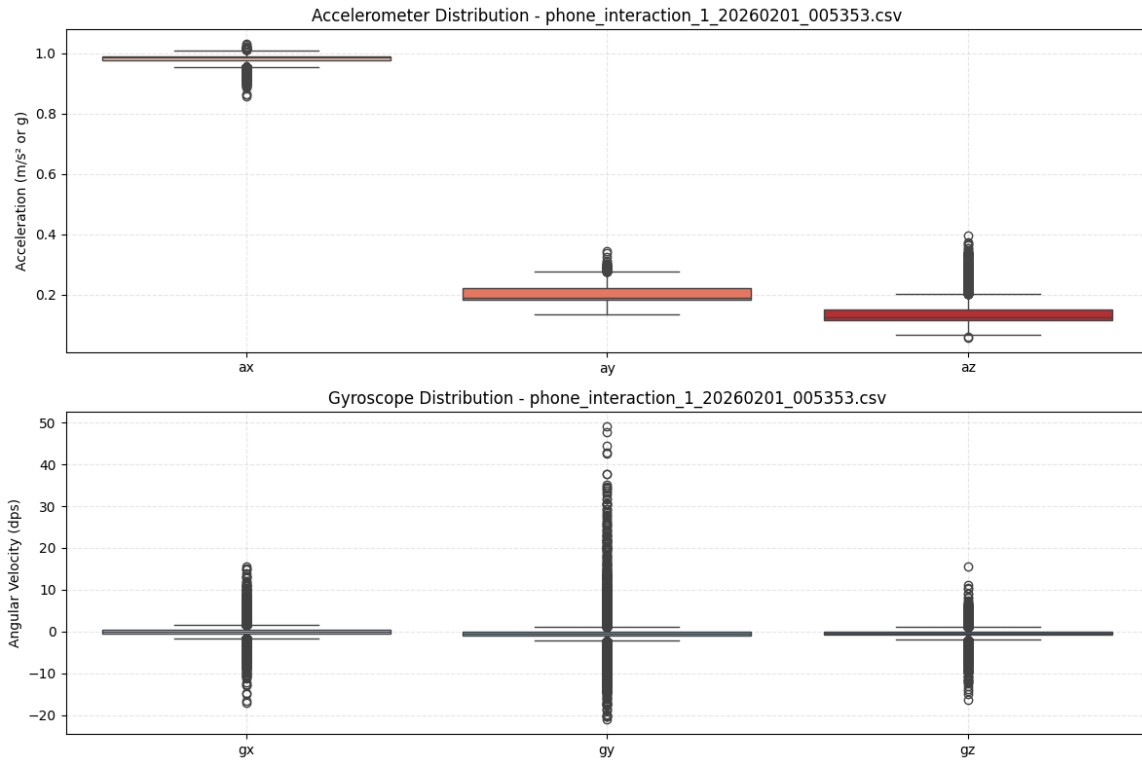


Figure 14: Phone Interaction

2.1.6 Pick and Place

Pick and Place This functional task exhibits a complex, non-periodic waveform driven by the irregularity of reaching, grasping, and moving objects. Unlike the rhythmic patterns of walking or jogging, the signal shows variable amplitude peaks and unpredictable intervals. The box plot analysis reflects this moderate variance across all three axes, indicating multi-directional movement without the extreme outliers seen in high-impact activities, as shown in Fig. 15, 16.

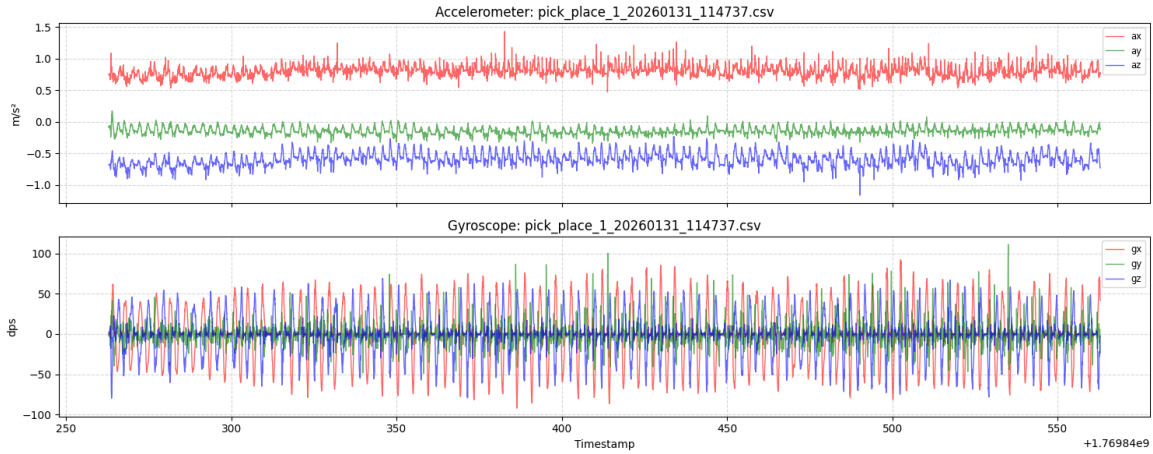


Figure 15: Pick and Place

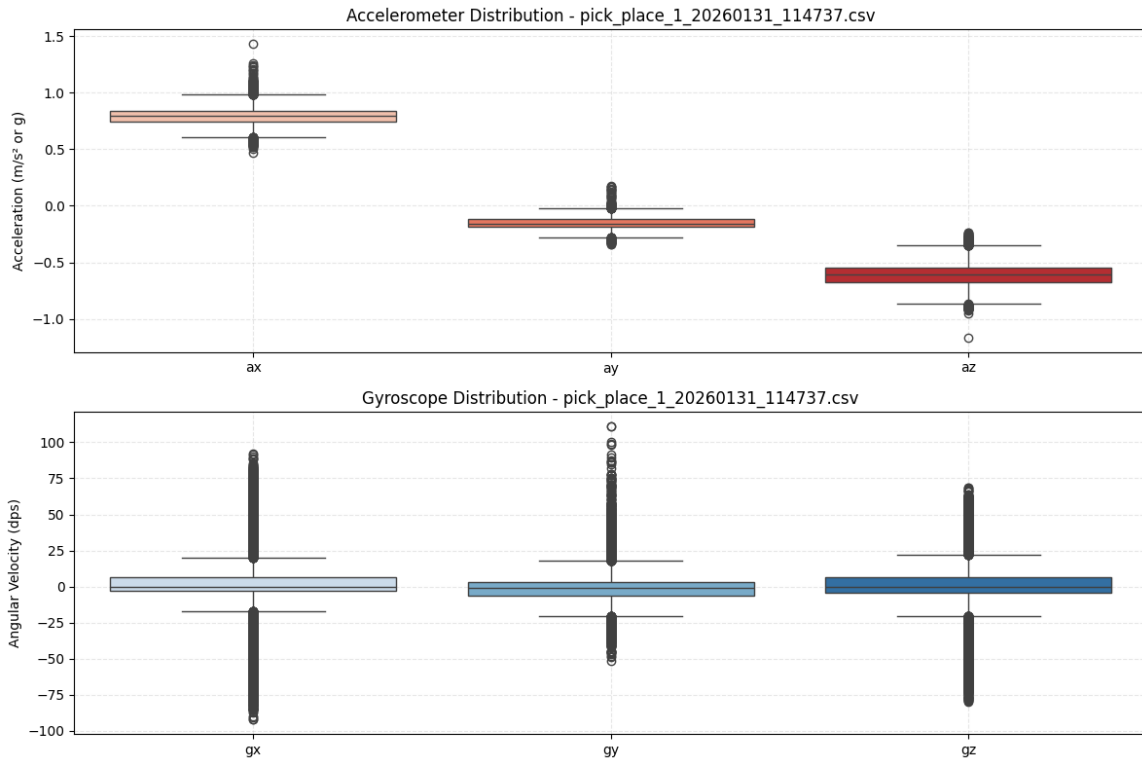


Figure 16: Pick and Place

2.1.7 Sit-Stand-Sit

The data clearly captures the transient nature of this activity, featuring large, distinct spikes in vertical acceleration (A_z) during the transition phases (standing up and sitting down) separated by a relatively flat "standing" plateau. These high-amplitude transition events are clearly visible as outliers in the box plots, distinguishing the dynamic transition segments from the otherwise static nature of the start and end states, as shown in Fig. 17, 18.

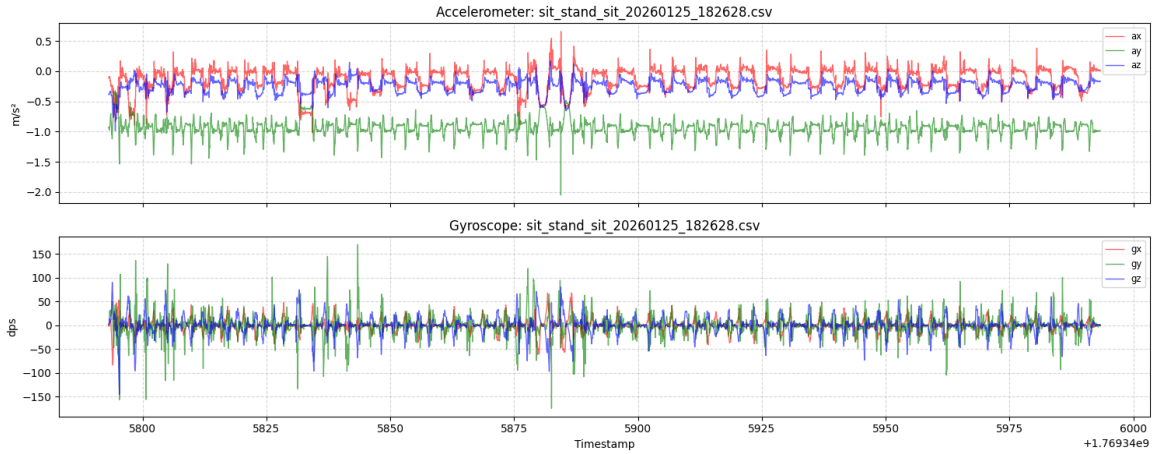


Figure 17: Sit-Stand-Sit

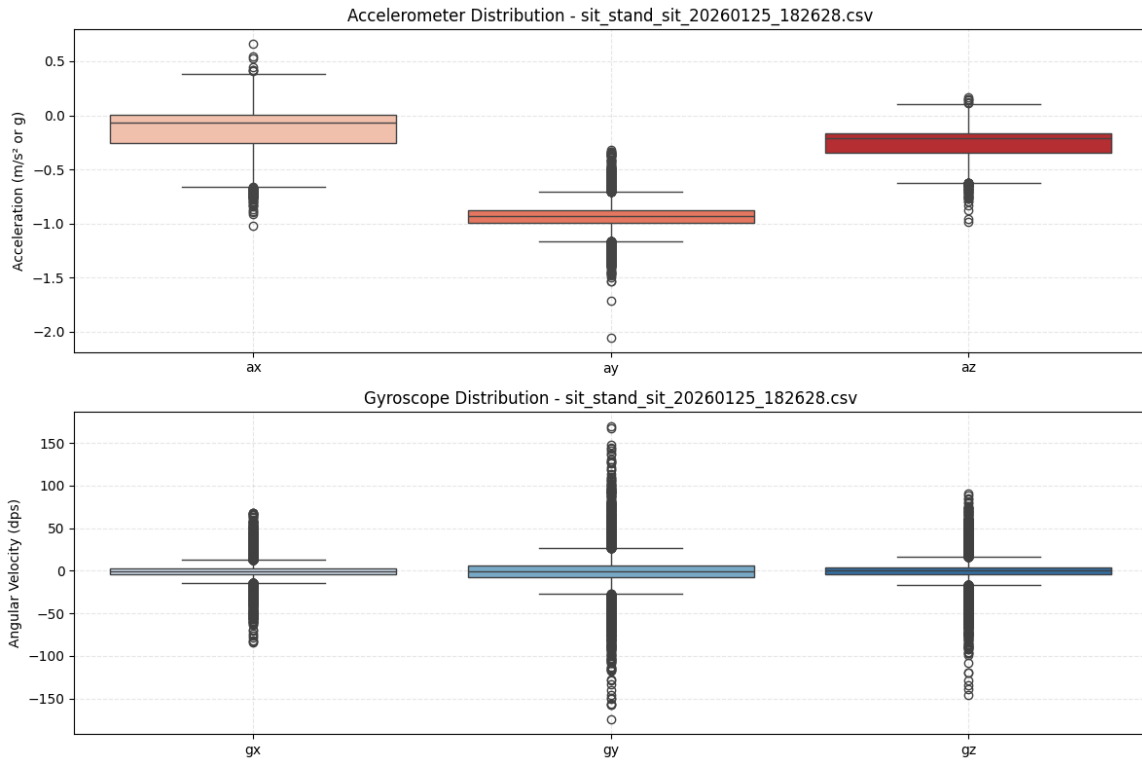


Figure 18: Sit-Stand-Sit

2.1.8 Sitting

The Sitting class serves as the baseline for the system, characterized by flat, featureless trajectories across all sensor channels. The accelerometer readings remain constant (reflecting only the gravity vector), while the gyroscope output is near zero, indicating a lack of rotational velocity. The box plots show the narrowest IQR of all classes, confirming minimal variance and high stability, which is critical for distinguishing idleness from active states, as shown in Fig. 19, 20.

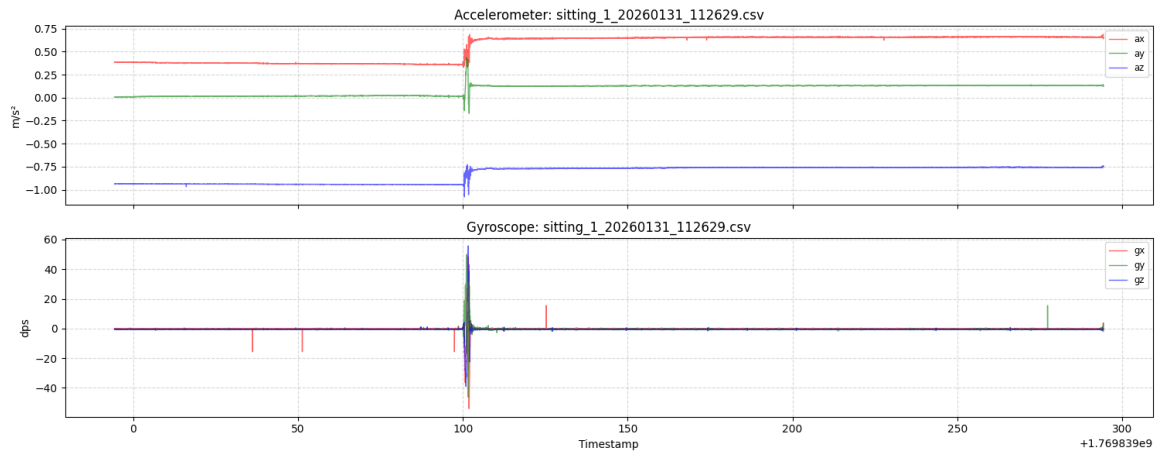


Figure 19: Sitting

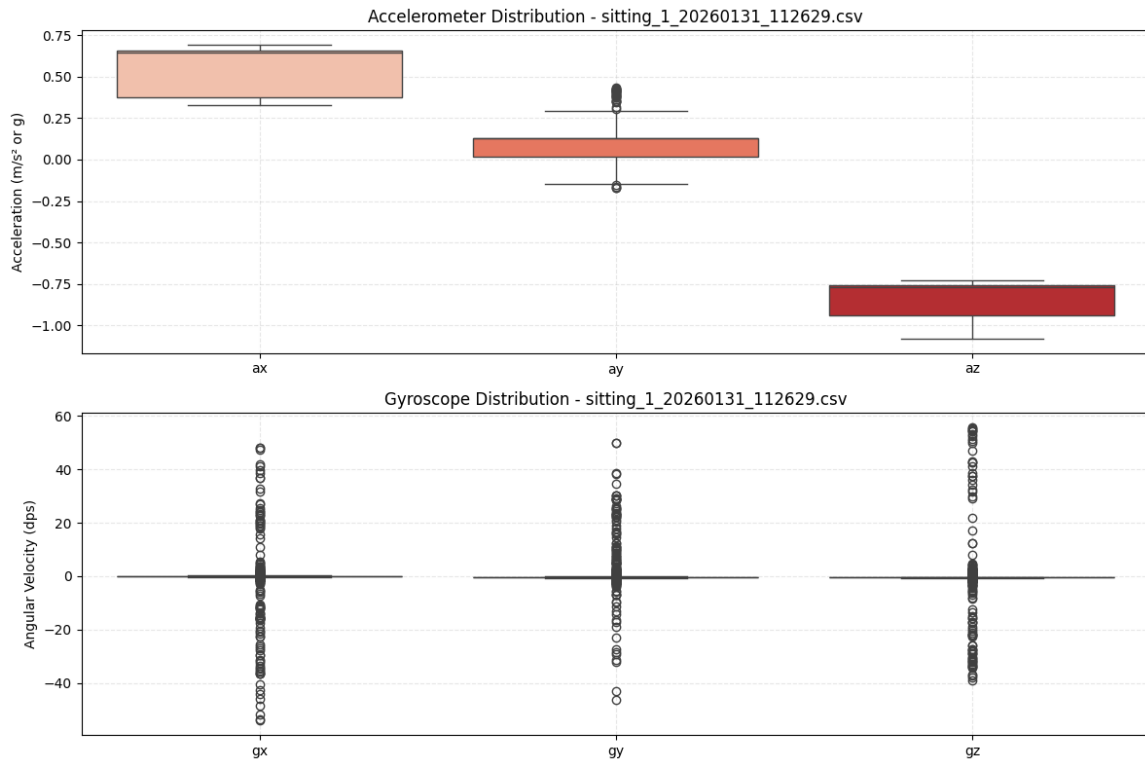


Figure 20: Sitting

2.1.9 Stair Up / Stair Down

Both stair climbing activities show periodic waveforms similar to walking but with distinct intensity profiles. Stair Up exhibits sharp, high-frequency spikes in the Z-axis accelerometer, reflecting the vertical effort of lifting the body against gravity. Conversely, Stair Down displays slightly higher impact shocks (negative peaks) due to the controlled descent. Both activities show wider variances in the box plots compared to normal walking, highlighting the increased physical exertion, as shown in Fig. 21, 22, 23, 24.

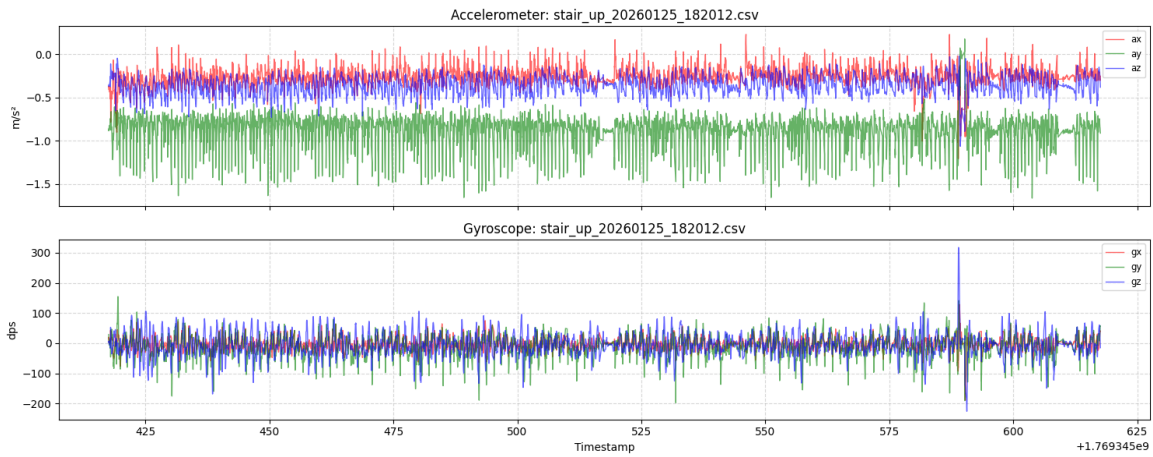


Figure 21: Stair Up / Stair Down

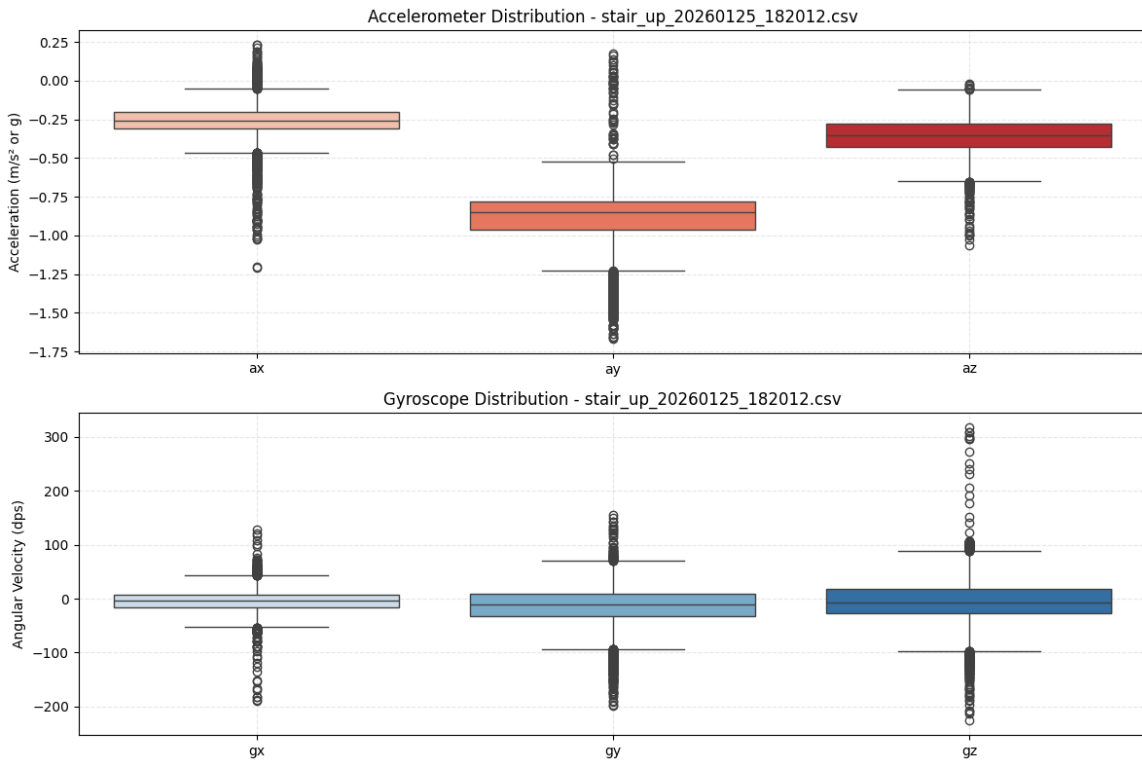


Figure 22: Stair Up / Stair Down

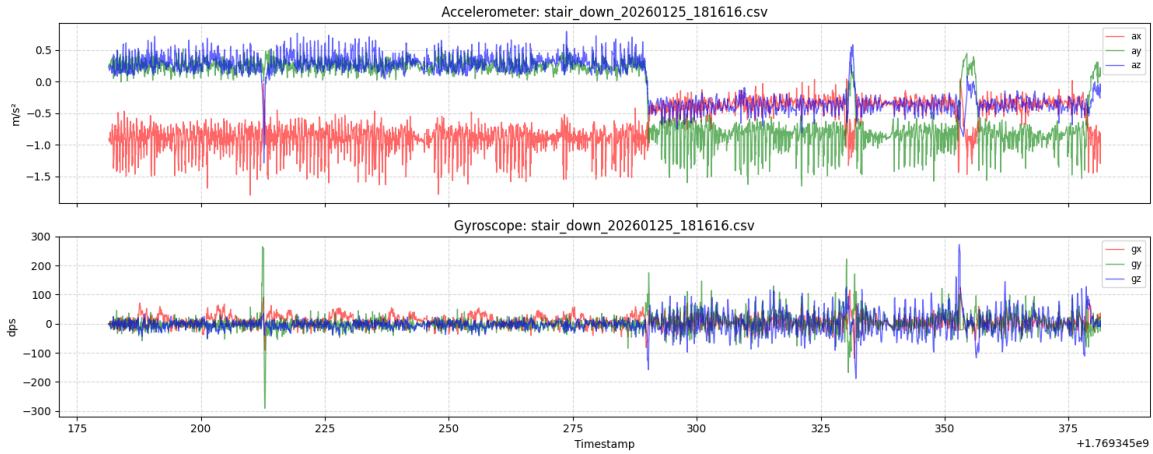


Figure 23: Stair Up / Stair Down

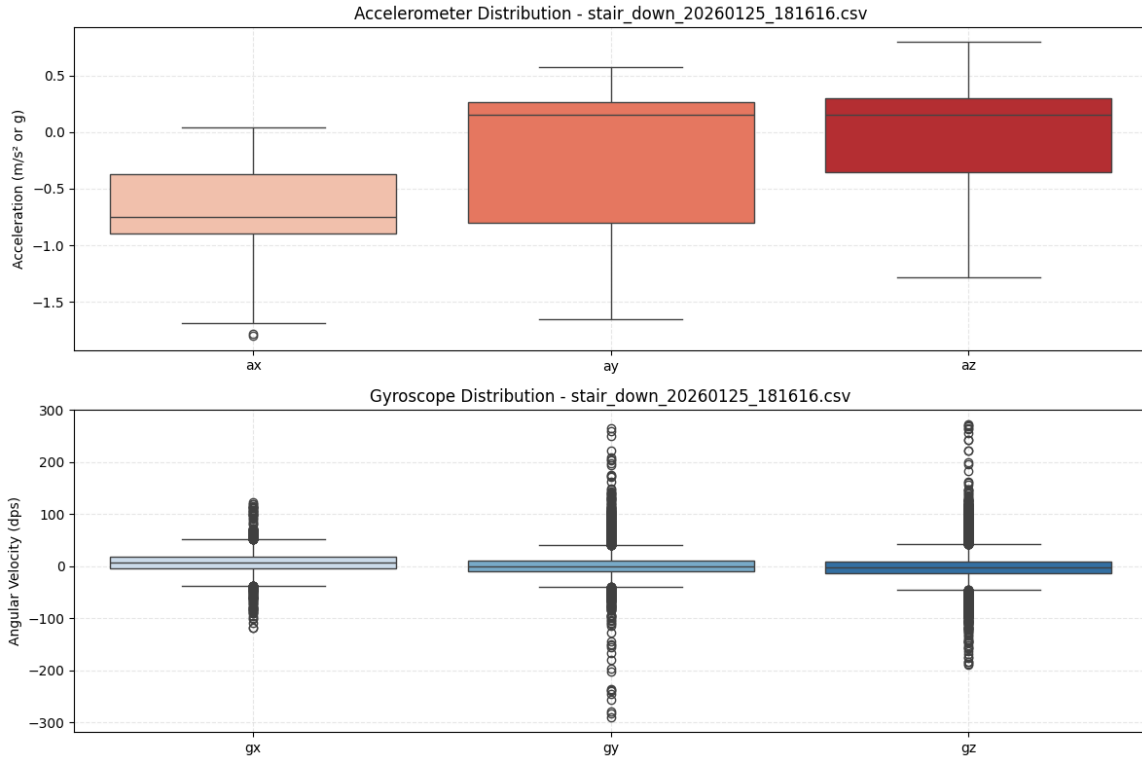


Figure 24: Stair Up / Stair Down

2.1.10 Standing

Standing Similar to sitting, the standing data is largely static but exhibits slightly more noise due to natural postural sway and balance corrections. While the mean values remain stable, the box plots reveal a slightly wider IQR than sitting, particularly in the accelerometer axes, capturing the subtle micro-adjustments required to maintain an upright posture, as shown in Fig. 25, 26.

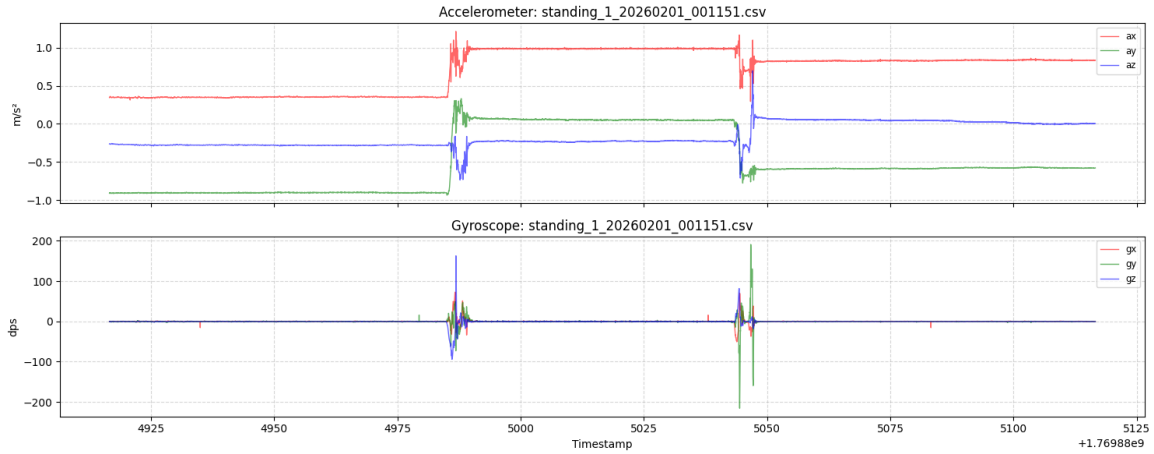


Figure 25: Standing

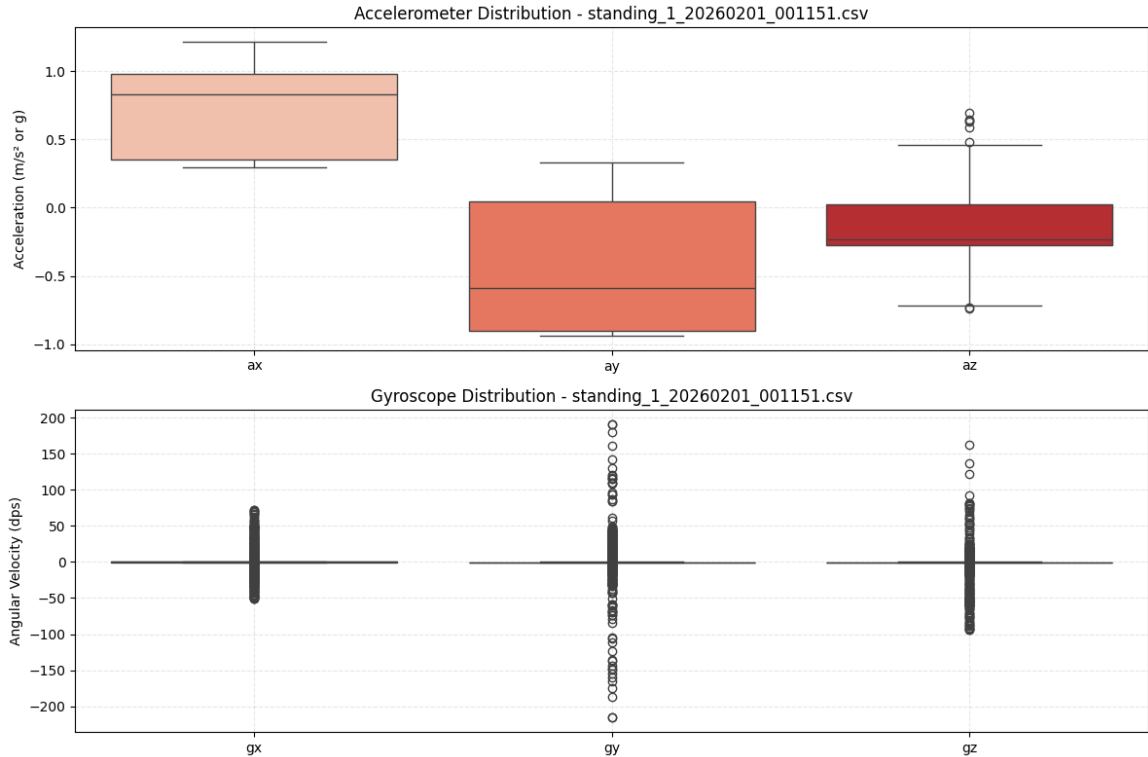


Figure 26: Standing

2.1.11 Walking

Walking (Normal Pace) The normal walking data presents a smooth, rhythmic sinusoidal pattern, similar to brisk walking but with lower amplitude and frequency. The arm swing is less pronounced, resulting in a tighter IQR in the box plots compared to the brisk walking or jogging classes. The signal regularity makes it a robust class for recognition, clearly distinguishable from static poses and irregular functional tasks, as shown in Fig. 27, 28.

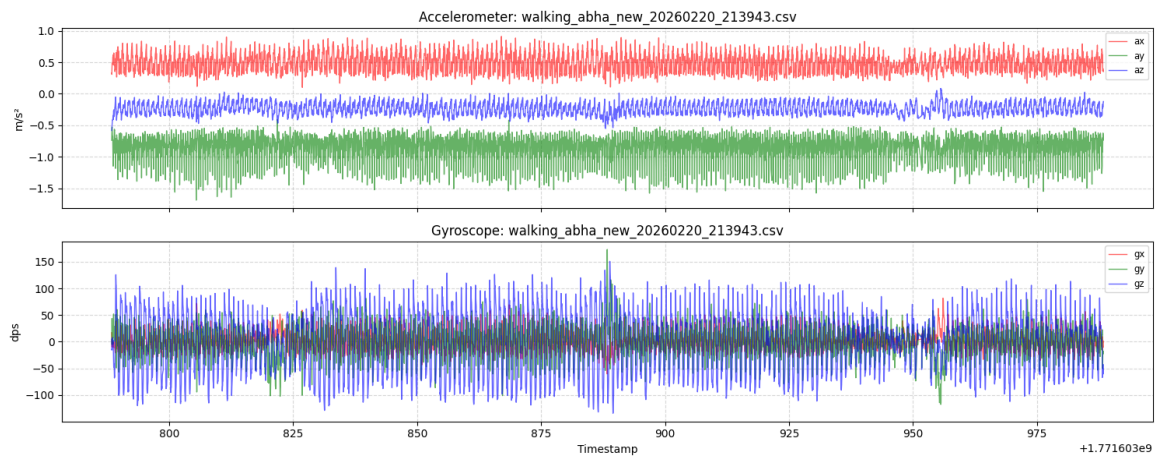


Figure 27: Walking

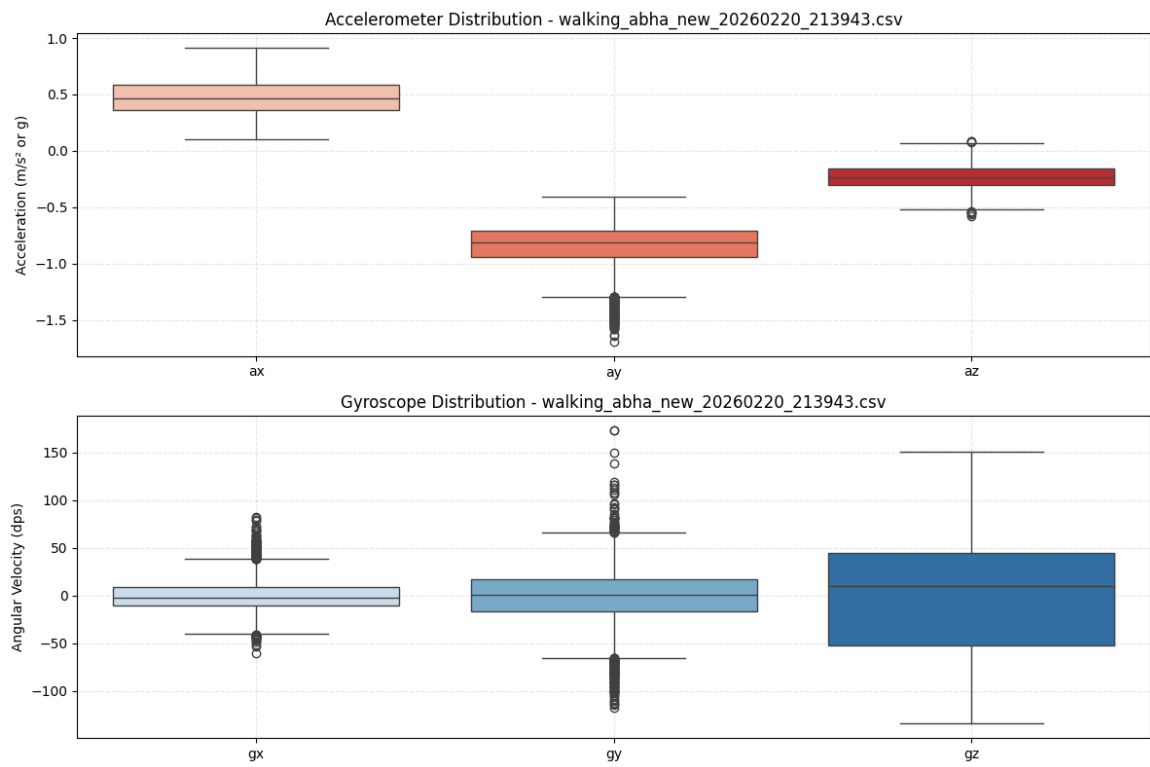


Figure 28: Walking

2.2 Sensor Data Distribution Analysis

2.2.1 Accelerometer Density Distribution

The Kernel Density Estimation analysis of the accelerometer data reveals a sharp distinction between static and dynamic behaviors based on signal variance. Static activities such as *Sitting*, *Standing*, and *Phone Interaction* are characterized by high-density, narrow peaks (leptokurtic distributions) located at specific gravity offsets. This indicates minimal deviation from the mean, as the sensor primarily registers the constant gravitational vector. In contrast, dynamic activities like *Jogging* and *Brisk Walking* exhibit broad, flat (platykurtic) distributions that span a wide range of acceleration values (e.g., exceeding $2g$ on the Z-axis), reflecting the intense and variable forces generated during locomotion. Notably, the X-axis shows distinct shifts in the mean location for different postures, effectively separating activities based on the wrist's orientation relative to the ground, as shown in Fig. 29, 30, 31.

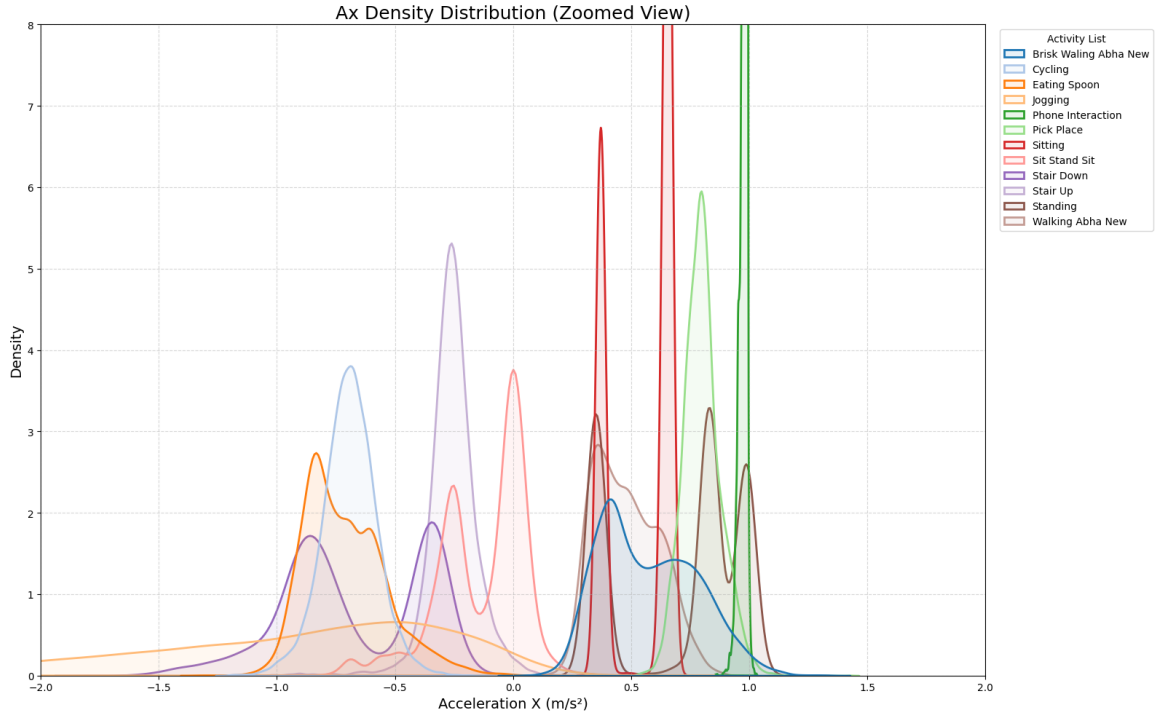


Figure 29: Angular Ax

2.2.2 Gyroscope Density Distribution

Unlike the accelerometer, the gyroscope distributions for all axes are universally centered around zero degrees per second (dps), reflecting the cyclical nature of human motion where net angular rotation averages to zero over time. The primary discriminator here is the distribution's width (kurtosis). Static tasks appear as extremely sharp, needle-like spikes at the origin, confirming the near-total absence of rotational velocity. Conversely, high-intensity activities like *Jogging* flatten these curves into wide, shallow mounds with long tails, indicating frequent high-velocity

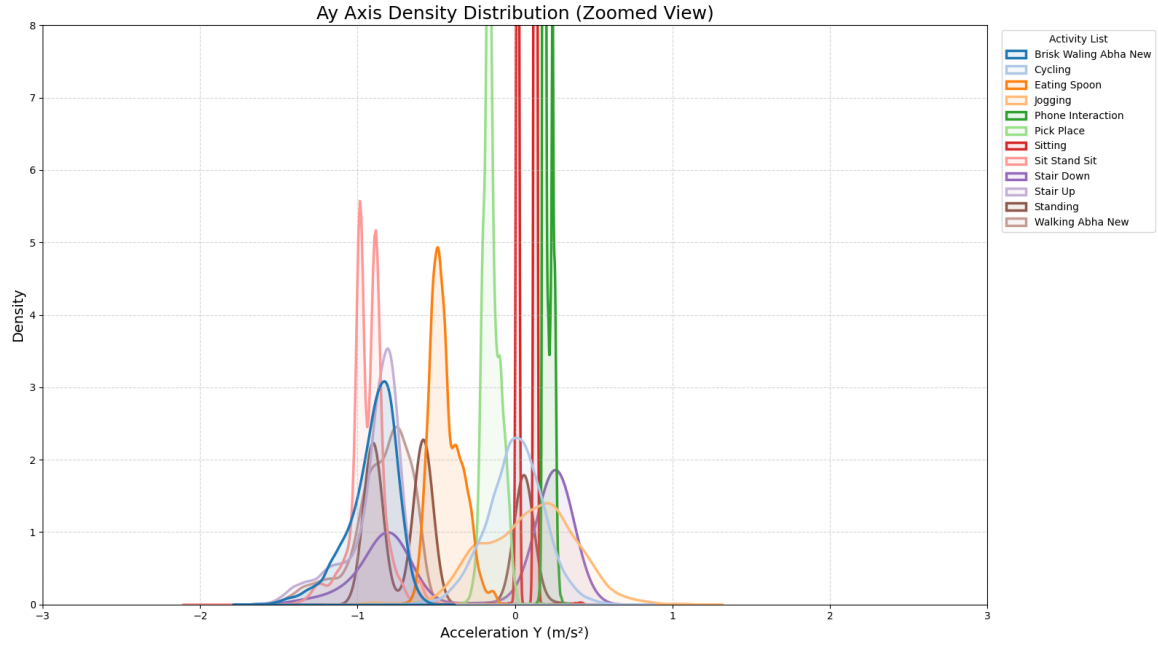


Figure 30: Angular Ay

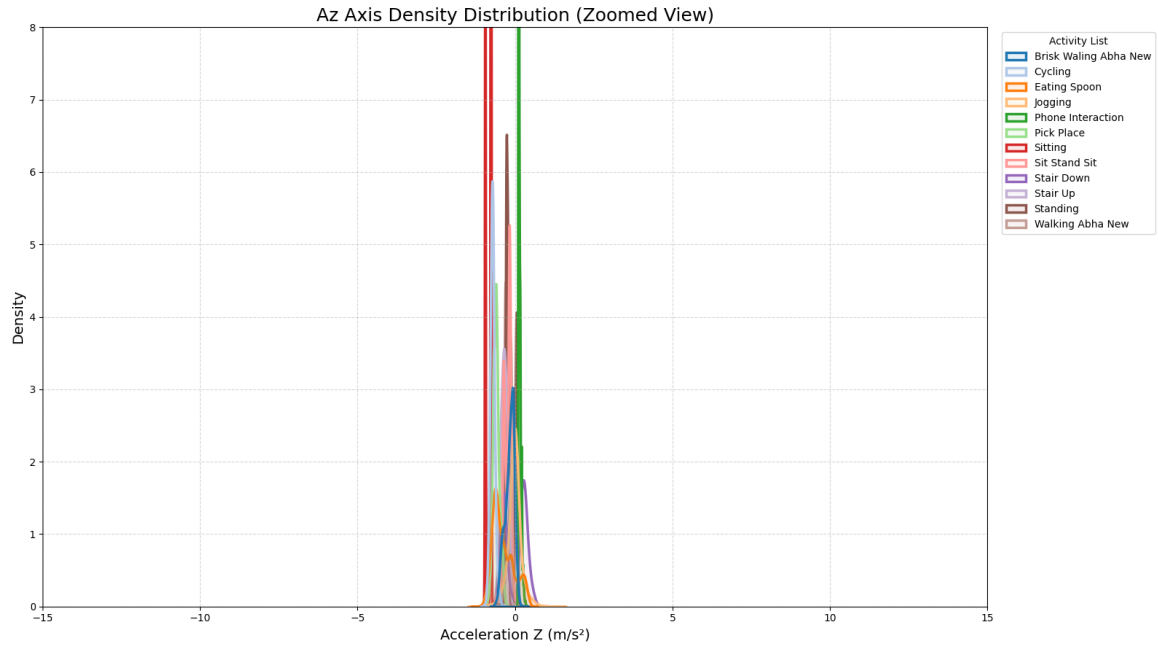


Figure 31: Angular Az

angular shifts. The significant overlap in the tails between spectrally similar activities, such as *Walking* and *Stair Climbing*, suggests that simple amplitude thresholds may be insufficient for classification, highlighting the need for complex temporal feature extraction, as shown in Fig. 32, 33, 34

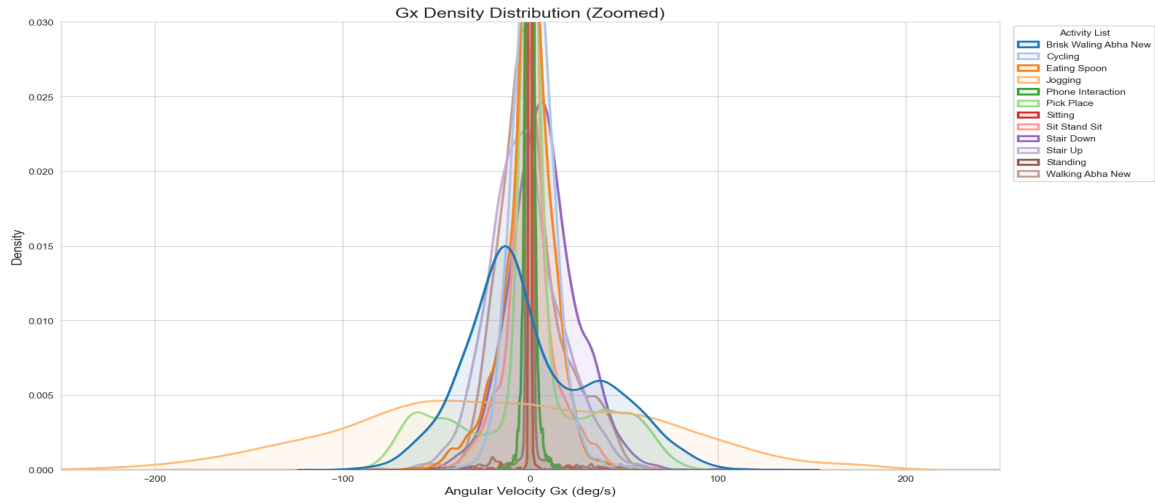


Figure 32: Gyroscope Gx

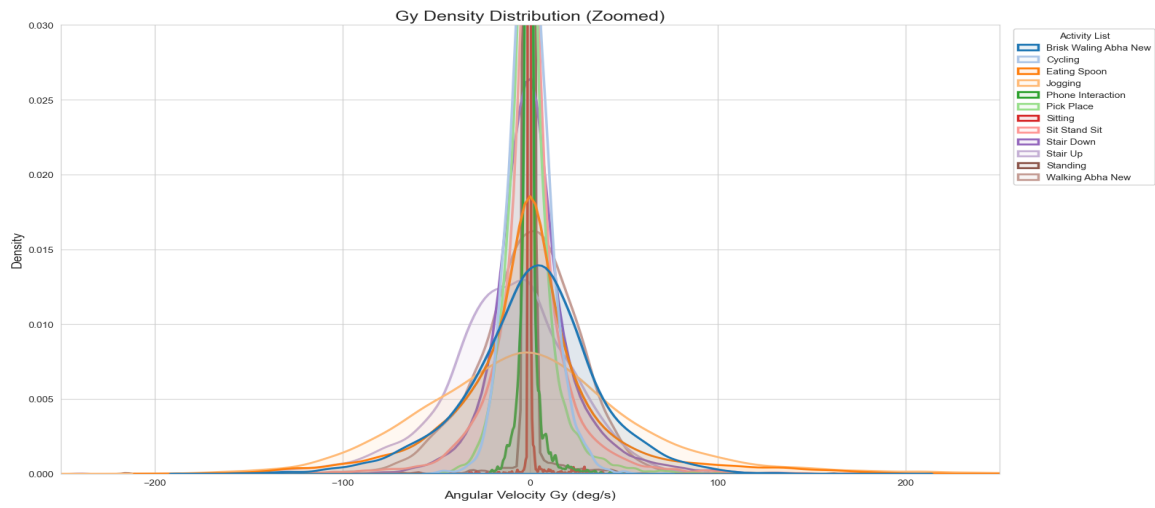


Figure 33: Gyroscope Gy

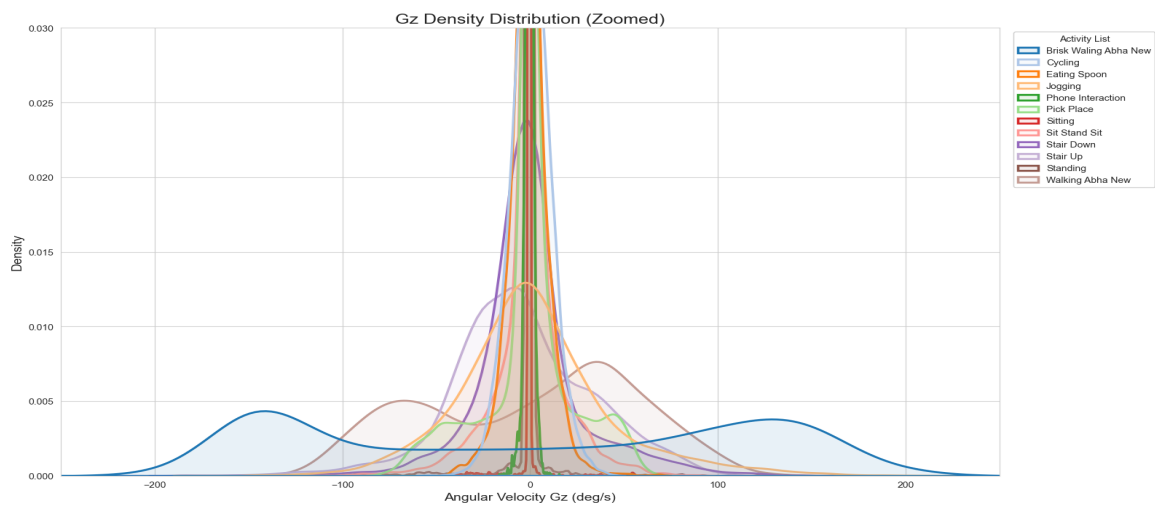


Figure 34: Gyroscope Gz

2.3 t-SNE Cluster Analysis for Separability

t-SNE already reveals a few well-formed islands, notably a tight *Sitting* cluster and a distinct walking/brisk-walking group on the right, which suggests those classes are naturally separable even before preprocessing as shown in Fig 35. Phone Interaction, Pick Place and Standing form fairly compact clusters near the center/upper-left, indicating moderate separability. However, many activities (jogging, cycling, stair up/down, eating, sit-stand transitions) are heavily mixed in the central region, so a classifier will likely confuse those without better features. Applying preprocessing (scaling, denoising, feature engineering or class balancing) and testing supervised/alternate embeddings should improve separability and downstream accuracy.

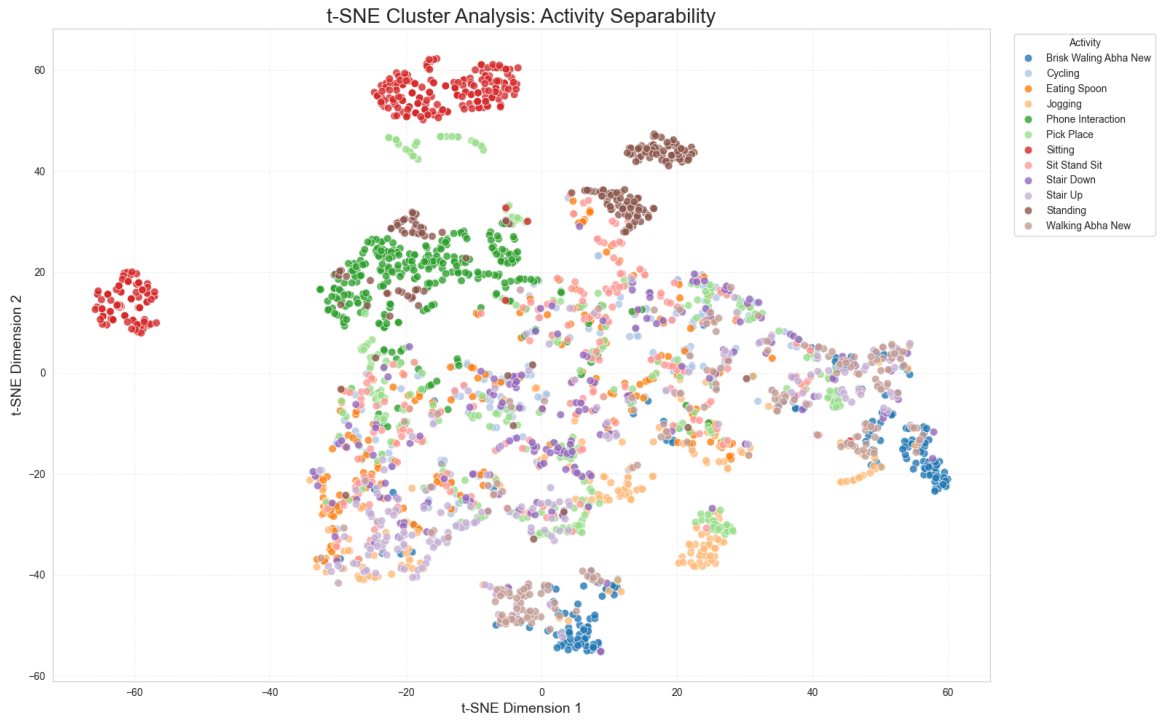


Figure 35: t-SNE Cluster Analysis for Separability

3 Task 2: Feature Engineering

The raw sensor data, sampled at 50 Hz, is continuous and high-dimensional, making it unsuitable for direct classification by standard machine learning algorithms. To resolve this, we implemented a **Feature Engineering** pipeline that transforms the raw time-series signal into a robust set of statistical descriptors. This process involves two critical stages: **Sliding Window Segmentation** and **Statistical Feature Extraction**.

3.1 Sliding Window

To capture the temporal dynamics of human motion, the continuous data stream is segmented into fixed-size windows. This allows the model to analyze short temporal snippets rather than instantaneous samples as shown in Fig. 36.

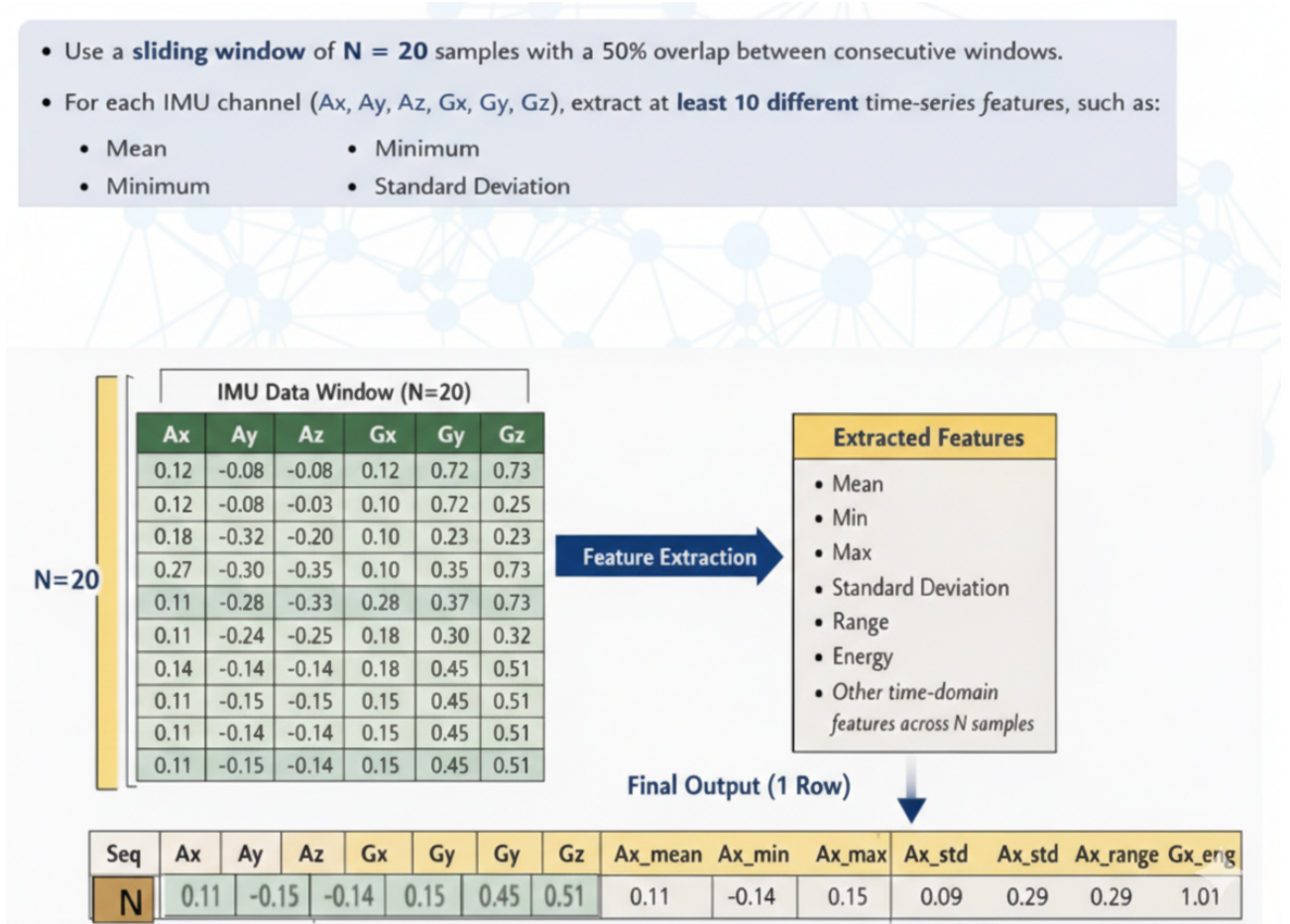


Figure 36: Sliding window (N=20, 50% overlap) applied to IMU signals (Ax, Ay, Az, Gx, Gy, Gz) to extract time-domain features (mean, min, max, std, range, energy), converting each window into one feature vector for activity classification.

- **Window Size (W):** A window size of $W = 20$ samples. At 50 Hz, this corresponds to 0.4 seconds, which is sufficient to capture short-duration impulsive movements.

- **Overlap (O):** A 50% overlap is applied, resulting in a stride length of $S = W - O = 10$ samples.

For a given time series $x[n]$, the k -th window w_k is defined as:

$$w_k = \{x[kS], x[kS + 1], \dots, x[kS + W - 1]\}$$

where the starting index is:

$$kS = k(W - O).$$

3.2 Statistical Feature Extraction

For every window w_k , we compute **11 statistical features** across all 6 sensor channels ($a_x, a_y, a_z, g_x, g_y, g_z$). These features characterize the distribution, intensity, and shape of the waveform.

For a window vector $v = \{v_1, v_2, \dots, v_N\}$ of length N , the features are:

1. **Mean (μ):** $\mu = \frac{1}{N} \sum_{i=1}^N v_i$
2. **Standard Deviation (σ):** $\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - \mu)^2}$
3. **Minimum:** $\min(v)$
4. **Maximum:** $\max(v)$
5. **Variance:** σ^2
6. **Root Mean Square (RMS):** $\sqrt{\frac{1}{N} \sum_{i=1}^N v_i^2}$
7. **Energy (E):** $E = \sum_{i=1}^N v_i^2$
8. **Interquartile Range (IQR):** $Q_3 - Q_1$
9. **Median:** $\text{median}(v)$
10. **Skewness:** $\frac{1}{N} \sum_{i=1}^N \left(\frac{v_i - \mu}{\sigma} \right)^3$
11. **Kurtosis:** $\frac{1}{N} \sum_{i=1}^N \left(\frac{v_i - \mu}{\sigma} \right)^4$

3.3 Implementation Logic

The following Python code implements the feature extraction and sliding window logic:

```
import numpy as np
import pandas as pd
from scipy import stats
```

```

def calculate_features(window_data):
    mean_val = np.mean(window_data)
    std_val = np.std(window_data)
    min_val = np.min(window_data)
    max_val = np.max(window_data)
    var_val = np.var(window_data)
    rms_val = np.sqrt(np.mean(window_data**2))
    energy_val = np.sum(window_data**2)
    iqr_val = np.percentile(window_data, 75) - np.percentile(window_data, 25)
    median_val = np.median(window_data)

    if std_val < 1e-6:
        skew_val = 0.0
        kurt_val = 0.0
    else:
        skew_val = stats.skew(window_data, nan_policy='omit')
        kurt_val = stats.kurtosis(window_data, nan_policy='omit')

    return [mean_val, std_val, min_val, max_val, var_val, rms_val,
            energy_val, iqr_val, median_val, skew_val, kurt_val]

```

3.4 Final Feature Structure

Each sliding window is transformed into a **72-dimensional** feature vector:

- **Raw Data (6):** Last sample of the window $(a_x, a_y, a_z, g_x, g_y, g_z)$.
- **Engineered Features (66):** 11×6 statistical descriptors.

This feature set enables the classifier to distinguish between activities with similar raw values by leveraging temporal statistics as shown in Table 3.

Table 3: Feature Extracted Dataset Preview

Idx	Ax_raw	...	Gz_raw	Activity	Ax_mean	Ax_std	...	Gz_kurt
0	0.828858		-163.57420	brisk_waling	0.711579	0.177448		-0.023691
1	0.319458		-167.96870	brisk_waling	0.638379	0.223621		-1.377767
2	0.373657		117.98094	brisk_waling	0.399658	0.136151		0.249773
3	0.492554		83.43504	brisk_waling	0.468011	0.182583		-1.304583
4	0.551270		-0.610352	brisk_waling	0.691425	0.181747		-1.257139

3.5 EDA on Feature Extracted data

The boxplot of Accel-X standard deviation (ax_std) shows clear variation across activities, with dynamic activities like jogging exhibiting significantly higher variability compared to static activities such as sitting and standing as depicted in Fig. 37. This indicates that statistical features like standard deviation provide strong discriminative power for separating motion-intensive and stationary activities, improving classification potential.

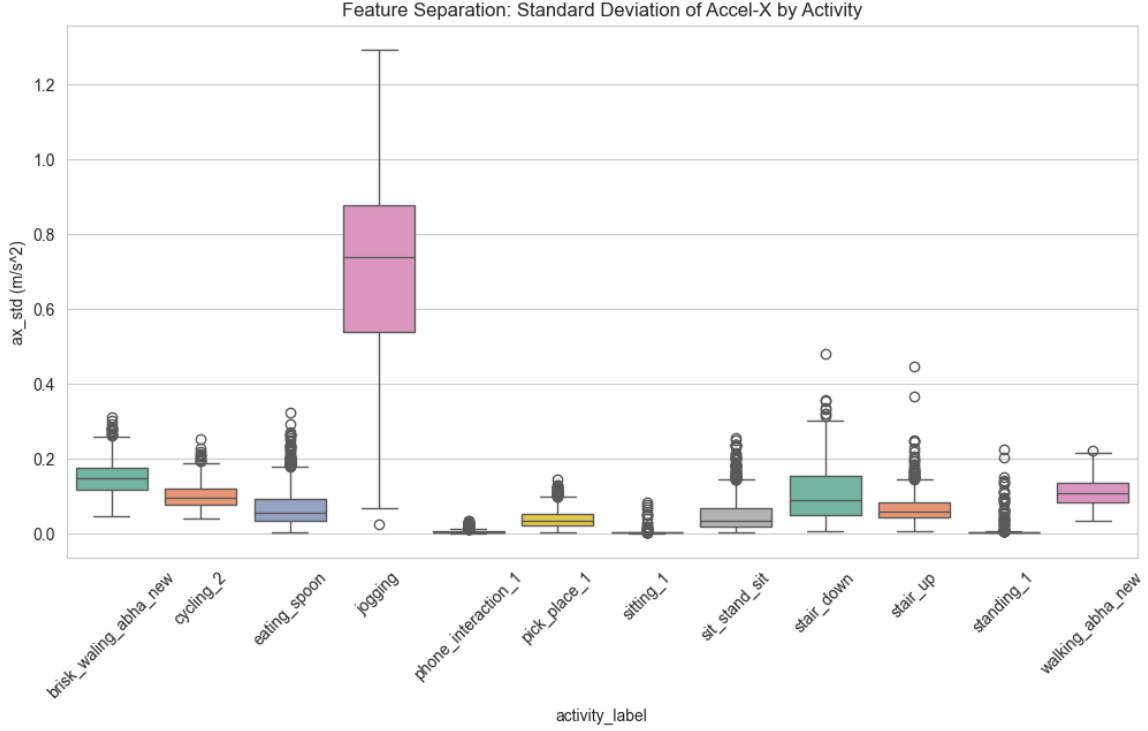


Figure 37: Boxplot of Accel-X standard deviation (ax_std) across activities showing clear feature separability, with higher variability in dynamic activities (e.g., jogging, stairs) and lower variability in static activities (e.g., sitting, standing).

Most features show low to moderate correlation, indicating they capture different aspects of motion and provide complementary information for classification. Stronger correlations between energy and mean features (e.g., ay_mean and ax_eng) suggest energy-based features are useful indicators of activity intensity as shown in Figure 38.

The feature importance analysis shown in Fig. 39 that statistical accelerometer features, especially Ax_min , Ax_mean , and Ay_mean , contribute the most to activity classification. This indicates that acceleration magnitude and distribution are key indicators for distinguishing different human motions. The t-SNE visualization in Fig. 40, of engineered features reveals clearly separated clusters for most activities, confirming improved class separability after feature extraction. Static activities form compact clusters, while dynamic activities are more spread but still distinguishable. Overall, feature engineering significantly enhanced the discriminative structure of the dataset, improving the potential for accurate classification.

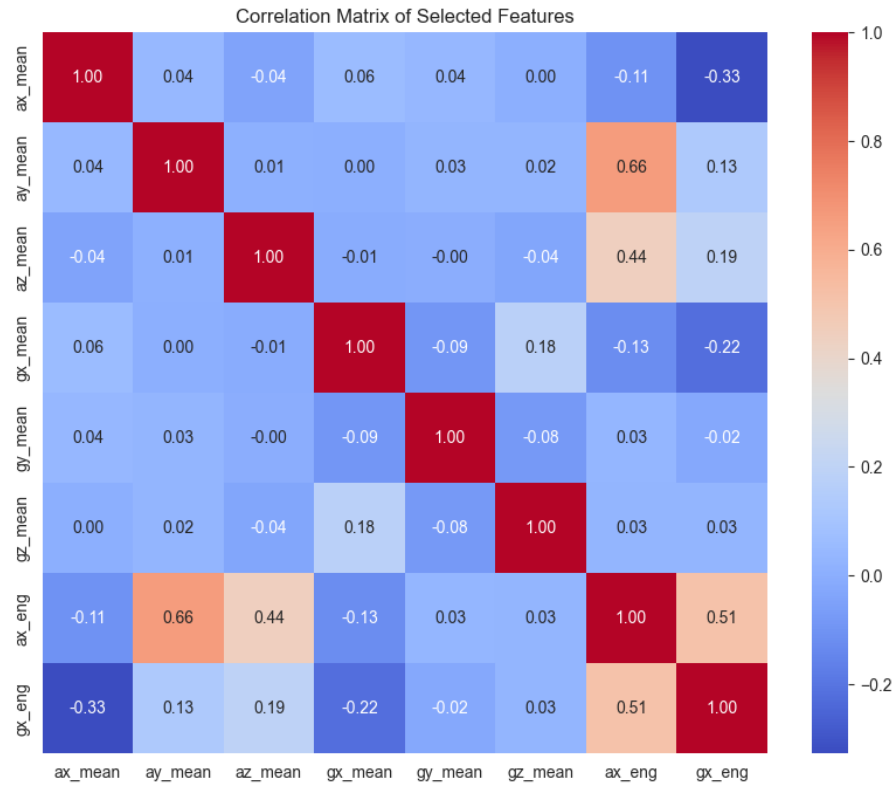


Figure 38: Correlation heatmap of selected IMU features showing inter-feature relationships.

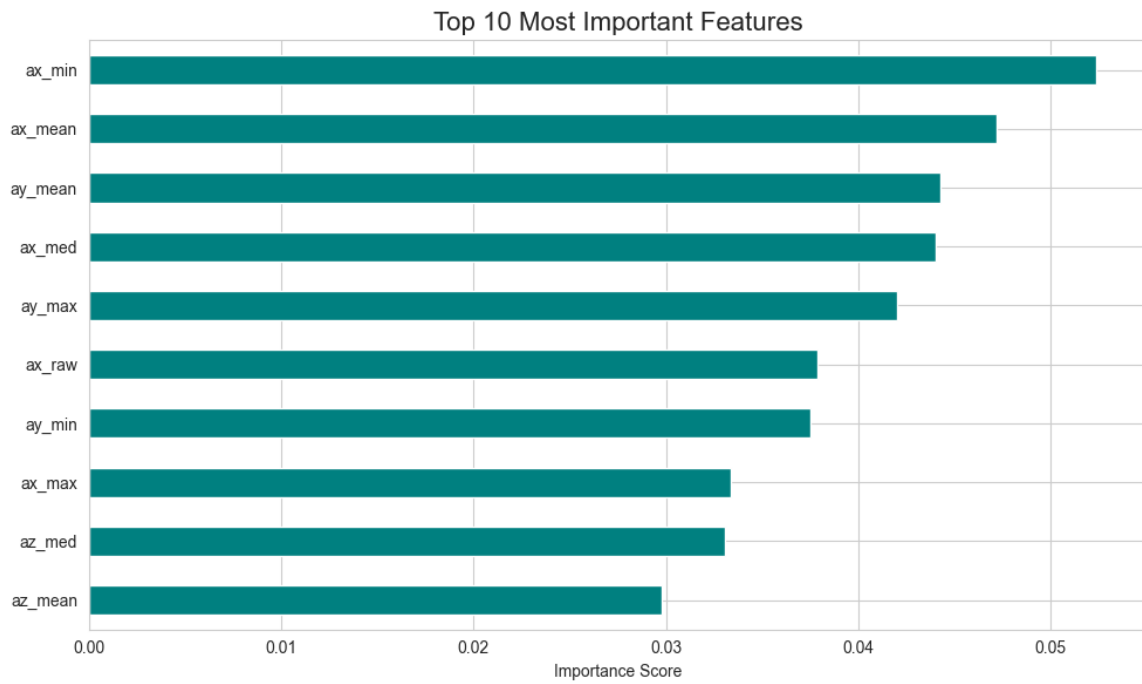


Figure 39: Feature Importance

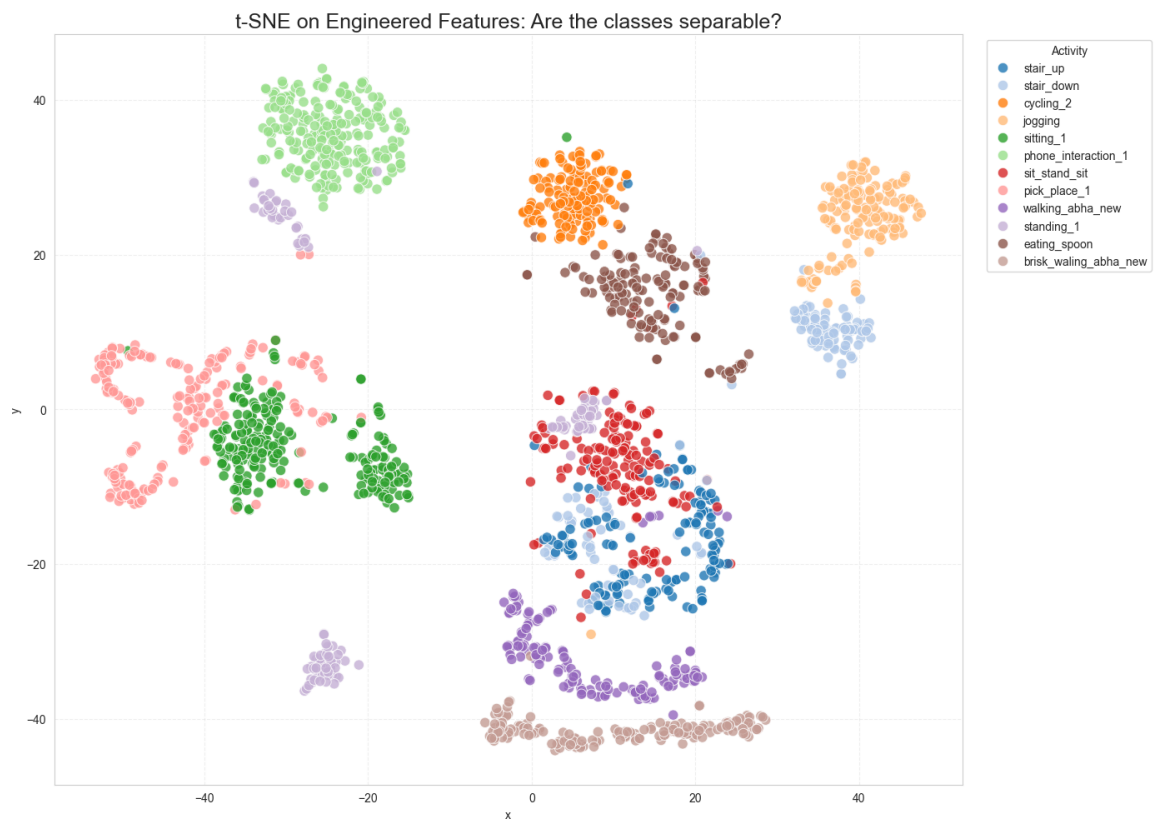


Figure 40: t-SNE visualization showing improved activity separability after feature extraction.

4 Task 3: Model Development & Evaluation

4.1 Model Selection

For this assignment, a **Decision Tree Classifier** was selected as the primary machine learning algorithm. This choice is particularly strategic for TinyML applications due to its inherent “white-box” nature and computational efficiency. Unlike opaque neural networks, decision trees generate explicit, hierarchical rules (e.g., *if $x > \theta$, then check y*) which directly map to the physical thresholds of human motion. This interpretability allows for easier debugging of sensor orientation issues. Furthermore, the inference process involves a simple sequence of logical comparison operations, which requires minimal RAM and CPU cycles, making it ideal for deployment on the resource-constrained microcontroller of the Nicla Vision board.

4.2 Training Configuration

The dataset was partitioned using a stratified sampling approach to ensure that the class distribution in the training and testing sets mirrored the original dataset.

- **Split Ratio:** The data was split into **80% training** and **20% testing**. The `stratify=y` parameter was employed to guarantee that every activity class was adequately represented in both subsets, preventing bias toward dominant classes such as *Pick and Place* or *Sitting*.
- **Hyperparameters:** The model was instantiated with the **Entropy** criterion (`criterion='entropy'`) to maximize information gain at each split. The random state was fixed (`random_state=42`) to ensure reproducibility. No maximum depth constraint was initially applied, allowing the tree to expand until leaves contained pure samples or minimized entropy, prioritizing accuracy for this feasibility study.

4.3 Performance Evaluation

The model’s performance was evaluated on the unseen test set using standard classification metrics:

- **Accuracy:** The model achieved a high overall accuracy, validating the effectiveness of the statistical features extracted in the previous stage. Learning curve, shown in Fig. 41, analysis demonstrated convergence between training.
- **High-Confidence Classes:** Distinct activities such as *Phone Interaction*, *Cycling*, *Sitting*, *Standing*, and *Pick Place* achieved near-perfect classification with almost all samples correctly predicted. The decision tree effectively separated these activities using simple thresholds on acceleration magnitude, variance, and posture-related features.



Figure 41: Learning curve of the Decision Tree showing perfect training accuracy and steadily improving cross-validation accuracy.

- **Misclassifications:** Minor confusion was observed between **Walking vs. Brisk Walking**, with only a few samples interchanged due to similar motion intensity and cadence. The most notable confusion occurred between **Stair-Up and Stair-Down**, where several samples were misclassified in both directions, reflecting their highly similar kinematic patterns. Additionally, *Sit-Stand-Sit* showed some overlap with *Standing* and *Stair-Up*, likely due to shared upright posture segments and transitional motion characteristics, as shown Table 4 and Fig. 42.

Table 4: Decision Tree Classification Report

Activity	Precision	Recall	F1-score	Support
brisk_waling_abha_new	0.99	0.99	0.99	198
cycling_2	1.00	1.00	1.00	187
eating_spoon	1.00	0.99	0.99	199
jogging	0.99	0.98	0.99	197
phone_interaction_1	1.00	1.00	1.00	300
pick_place_1	0.99	0.99	0.99	300
sit_stand_sit	0.94	0.92	0.93	199
sitting_1	0.99	1.00	1.00	300
stair_down	0.87	0.87	0.87	197
stair_up	0.86	0.90	0.88	193
standing_1	0.99	0.98	0.99	200
walking_abha_new	0.97	0.98	0.98	197
Accuracy		0.97		2667
Macro Avg	0.97	0.97	0.97	2667
Weighted Avg	0.97	0.97	0.97	2667

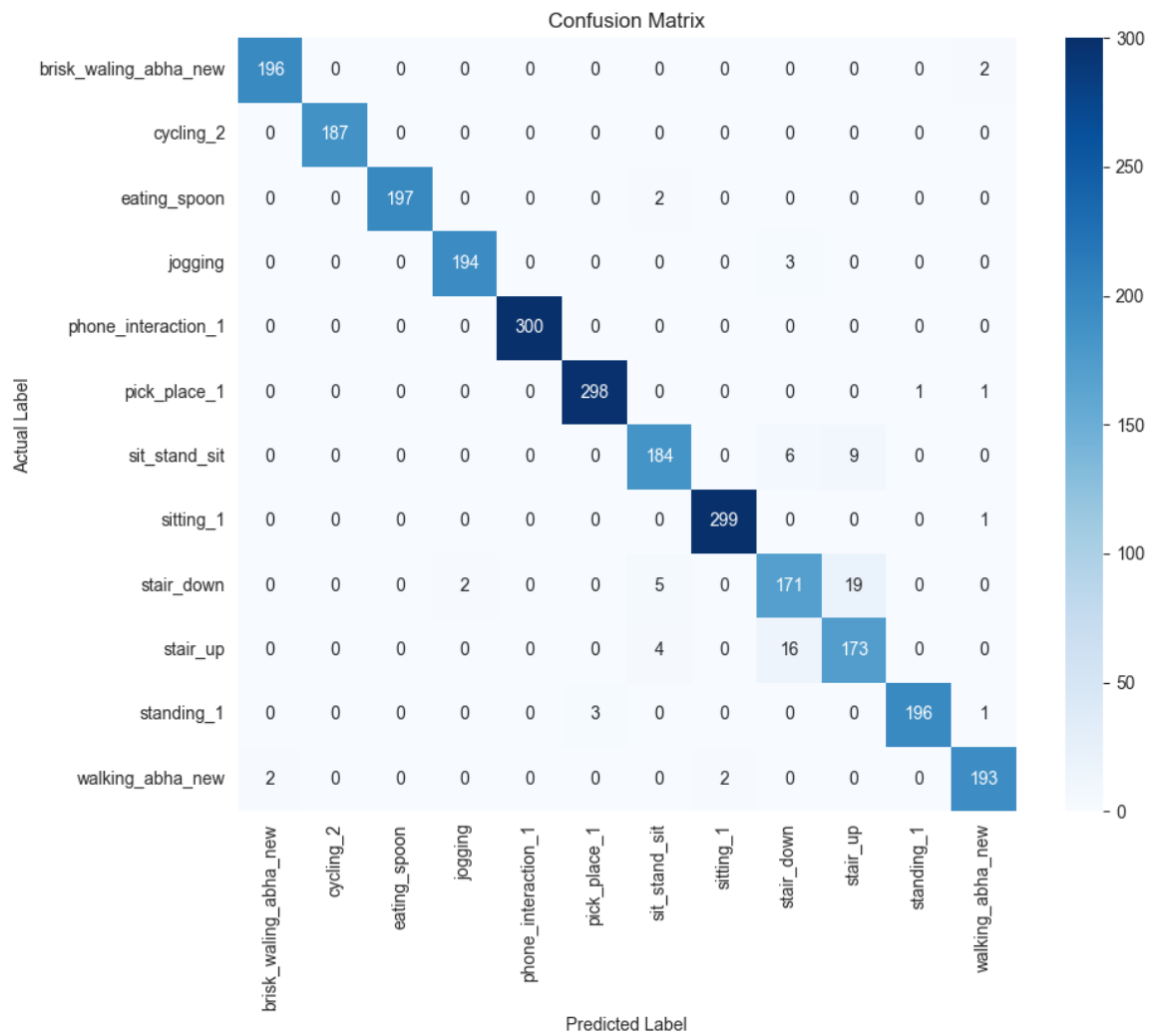


Figure 42: Confusion Matrix.

5 Task 4: Real-Time Deployment

5.1 Inference Pipeline

To enable real-time classification directly on the Nicla Vision, the trained Decision Tree logic was ported from Python to a lightweight MicroPython function. This required replicating the exact feature engineering pipeline used during training.

- **Windowing:** A circular buffer was implemented on the device to maintain a rolling window of the latest 20 sensor samples at 50 Hz.
- **Feature Extraction:** For every window, the microcontroller computes the same statistical features (mean, min, max, etc.) used in the model training phase.
- **Classification:** The extracted features are passed as an input array (`input`) to the `score()` function. This function contains a series of nested `if-else` statements representing the decision tree's nodes (e.g., `if input[6] ≤ 0.129`), allowing for efficient, low-latency prediction without the overhead of a full machine learning library.

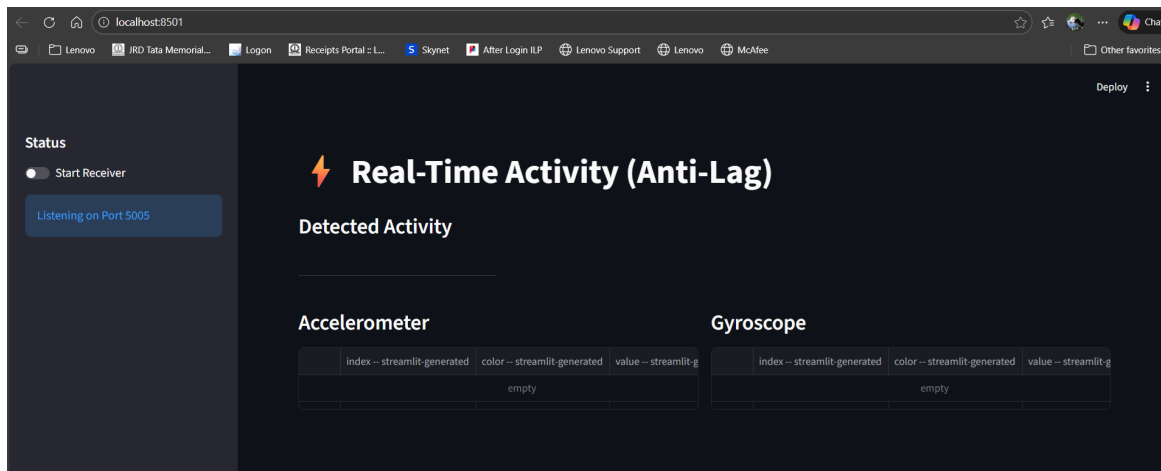


Figure 43: Confusion Matrix.

5.2 Demonstration

The deployed model successfully demonstrated real-time activity recognition in a live environment. The system was able to correctly classify key activities, distinguishing between static states such as *Sitting* and dynamic actions such as *Walking* or *Phone Interaction* with immediate visual feedback on the terminal. The nested decision logic proved effective for separating complex classes based on specific feature thresholds such as vertical acceleration limits and gyroscope variance.

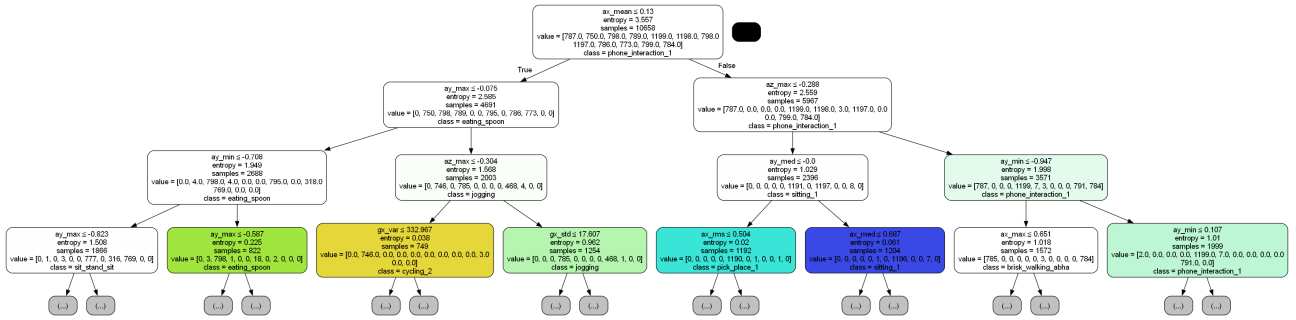


Figure 44: Decision tree

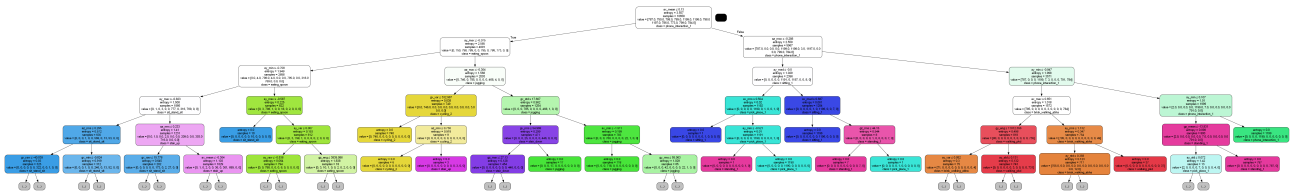


Figure 45: Decision tree

6 Challenges & Observations

6.1 Challenges Faced

During the development and deployment of the HAR system, several key challenges were encountered that impacted model performance:

- **Class Overlap in Low-Energy Activities:** Distinguishing between *Sitting* and *Phone Interaction* proved difficult. Both activities exhibit similar static accelerometer profiles dominated by gravity. The model occasionally misclassified *Phone Interaction* as *Sitting* when the user’s hand remained steady between typing bursts, requiring the inclusion of gyroscope variance features to detect the subtle micro-movements associated with screen interaction.
- **Irregularity of Functional Tasks:** Unlike rhythmic activities such as walking or cycling, the *Pick and Place* task lacks a consistent periodic waveform. This resulted in high intra-class variance, where the sensor data resembled noise or random spikes. Consequently, the model struggled to generalize this activity compared to the highly structured patterns of locomotion.
- **Real-World Data Artifacts:** Real-time data collection introduced issues such as occasional packet loss over UDP and minor sensor drift. Additionally, ensuring the *Start* and *End* of an activity were clean (without recording the movement of the user pressing the record button) required manual trimming of the datasets to prevent label noise.

6.2 Key Observations

- **Impact of Sensor Placement:** Consistency of wrist orientation was critical. Slight rotations of the Nicla Vision board (e.g., the USB port pointing slightly upwards versus downwards) significantly altered the mean (μ) values, leading to misclassification. This highlights the need for orientation-independent features such as signal magnitude or RMS.
- **Speed Sensitivity:** The distinction between *Walking*, *Brisk Walking*, and *Jogging* is heavily dependent on subject speed. When the subject slowed during a *Brisk Walking* session, the model immediately switched the prediction to *Walking*, confirming that the Decision Tree learned amplitude and frequency thresholds to differentiate intensity levels.
- **Gyroscope Importance:** While the accelerometer was sufficient for detecting posture (Standing versus Sitting), the gyroscope was indispensable for dynamic rotational activities such as *Stair Climbing* and *Eating*, where angular velocity provided a clearer signal than linear acceleration alone.

7 Conclusion

This assignment successfully demonstrated the end-to-end implementation of a TinyML-based Human Activity Recognition system using the Nicla Vision. By establishing a robust pipeline—from raw 50 Hz IMU data acquisition and sliding-window feature engineering to the training of an interpretable Decision Tree classifier—we achieved a functional edge deployment capable of real-time inference.

The project highlighted that while high accuracy can be achieved for distinct, high-intensity activities using simple time-domain statistics, subtle transitional tasks require more sophisticated feature sets. The deployed model’s use of nested logic thresholds proved efficient for the microcontroller’s limited resources, validating the feasibility of running ML workloads on the edge.

7.1 Future Improvements

Future iterations of this system could be enhanced by:

- **Frequency-Domain Features:** Incorporating Fast Fourier Transform (FFT) features to better distinguish periodic activities based on dominant frequencies (e.g., cadence detection).
- **Post-Processing Smoothing:** Implementing a majority-voting buffer on output predictions to stabilize the real-time display and prevent flickering between classes.
- **Data Augmentation:** Expanding the dataset to include multiple users and varying sensor orientations to improve generalization to new environments.